

Performance Analysis

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Detailed analysis of your performance in the AWS Certified Solutions Architect Associate - SAA C03 exam.

Answer Distribution

0

Correct

0

Incorrect

65

Unattempted

Domain Performance

Management & Governance

0%

Storage

0%

Database

0%

Security

0%

Compute

0%

Networking

0%

Monitoring

0%

Final Score

**0/65**

Passing score: 48

Time Taken

**130m 0s**

Total time: 130 minutes

Question Analysis

✓ Correct	0
✗ Incorrect	0
⊖ Unattempted	65
🔖 Reviewed	0

Question-wise Analysis

Expand All

Collapse All

Question 1

Not Attempted

Management & Governance



A company is building a solution that will report Amazon EC2 Auto Scaling events across all the applications in an AWS account. The company needs to use a serverless solution to store the EC2 Auto Scaling status data in Amazon S3. The company then will use the data in Amazon S3 to provide near-real-time updates in a dashboard. The solution must not affect the speed of EC2 instance launches. How should the company move the data to Amazon S3 to meet these requirements?

- ✓ Use an Amazon CloudWatch metric stream to send the EC2 Auto Scaling status data to Amazon Kinesis Data Firehose. Store the data in Amazon S3.

This solution is serverless and does not affect the speed of EC2 instance launches. CloudWatch metric streams continuously stream metrics to destinations such as Amazon S3 via Kinesis Data Firehose, which is designed for near-real-time data ingestion and delivery.

- ✗ Launch an Amazon EMR cluster to collect the EC2 Auto Scaling status data and send the data to Amazon Kinesis Data Firehose. Store the data in Amazon S3.

Using an Amazon EMR cluster introduces unnecessary complexity and overhead. It is not the most efficient serverless solution for this requirement.

- ✗ Create an Amazon EventBridge rule to invoke an AWS Lambda function on a schedule. Configure the Lambda function to send the EC2 Auto Scaling status data directly to Amazon S3.

Invoking a Lambda function on a schedule would not provide near-real-time updates and introduces delays in data availability. EventBridge is better suited to event-driven architectures, and scheduling defeats that purpose.

- ✗ Use a bootstrap script during the launch of an EC2 instance to install Amazon Kinesis Agent. Configure Kinesis Agent to collect the EC2 Auto Scaling status data and send the data to Amazon Kinesis Data Firehose. Store the data in Amazon S3.

Installing Kinesis Agent on each EC2 instance adds dependencies and management overhead, potentially impacting the speed of EC2 instance launches.

Overall Explanation:

The best approach is to use Amazon CloudWatch metric streams to send EC2 Auto Scaling status data to Amazon Kinesis Data Firehose, which then stores the data in Amazon S3.

This solution is serverless, provides near-real-time updates, and does not affect the speed of EC2 instance launches. Options B, C, and D introduce unnecessary complexity, delays, or potential performance impacts.

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Question 2

Not Attempted

Storage



A company has an application that places hundreds of .csv files into an Amazon S3 bucket every hour. The files are 1 GB in size. Each time a file is uploaded, the company needs to

convert the file to Apache Parquet format and place the output file into a different S3 bucket. Which solution will meet these requirements with the LEAST operational overhead?

- ⊗ Create an AWS Lambda function to download the .csv files, convert the files to Parquet format, and place the output files in an S3 bucket. Invoke the Lambda function for each S3 PUT event.

Using Lambda to download, convert, and upload may be problematic due to the file size (1GB) potentially exceeding Lambda's memory limits and execution time. It also requires managing the conversion process within the Lambda function.

- ⊗ Create an Apache Spark job to read the .csv files, convert the files to Parquet format, and place the output files in an S3 bucket. Create an AWS Lambda function for each S3 PUT event to invoke the Spark job.

While Spark can handle the conversion, managing a Spark cluster and invoking it with Lambda adds operational overhead.

- ⊗ Create an AWS Glue table and an AWS Glue crawler for the S3 bucket where the application places the .csv files. Schedule an AWS Lambda function to periodically use Amazon Athena to query the AWS Glue table, convert the query results into Parquet format, and place the output files into an S3 bucket.

Using Athena for conversion adds complexity and potentially higher costs. It is not the most efficient way to convert entire files.

- ✔ Create an AWS Glue extract, transform, and load (ETL) job to convert the .csv files to Parquet format and place the output files into an S3 bucket. Create an AWS Lambda function for each S3 PUT event to invoke the ETL job.

AWS Glue is a fully managed ETL service, making it the best option for this scenario. It handles the conversion process efficiently and reliably. Using Lambda to trigger the ETL job on each S3 PUT event provides near real-time conversion with minimal operational overhead. Glue can also be configured to trigger the ETL job without a Lambda function.

Overall Explanation:

AWS Glue is a fully managed ETL service designed for tasks like this. It can easily convert CSV files to Parquet format and place them in another S3 bucket. A Lambda function can be used to trigger the Glue job when a new file is uploaded, offering the least operational overhead compared to managing Spark clusters or handling conversions directly in Lambda. Although Glue can be set to trigger the ETL job directly, option D is the closest to

reality since S3 PUT events are immediate triggers which are most easily handled by Lambda.

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Question 3

Not Attempted

Database



A company is implementing new data retention policies for all databases that run on Amazon RDS DB instances. The company must retain daily backups for a minimum period of 2 years. The backups must be consistent and restorable. Which solution should a solutions architect recommend to meet these requirements?

- ☒ Create a backup vault in AWS Backup to retain RDS backups. Create a new backup plan with a daily schedule and an expiration period of 2 years after creation. Assign the RDS DB instances to the backup plan.

AWS Backup is a centralized backup management service suitable for retaining RDS backups. A backup vault is created, and a backup plan is defined with a daily schedule and a 2-year retention period for backups. RDS DB instances are assigned to this backup plan.

- ☐ Configure a backup window for the RDS DB instances for daily snapshots. Assign a snapshot retention policy of 2 years to each RDS DB instance. Use Amazon Data Lifecycle Manager (Amazon DLM) to schedule snapshot deletions.

While this option uses native RDS snapshot capabilities and DLM, option A is the better managed and centralized solution. DLM based snapshots are point in time and may not provide the desired level of consistency.

- ☐ Configure database transaction logs to be automatically backed up to Amazon CloudWatch Logs with an expiration period of 2 years.

CloudWatch Logs is not designed to provide the backup and restore functionality required for databases. It does not ensure the backups are consistent or provide an easy restore mechanism.

- ☐ Configure an AWS Database Migration Service (AWS DMS) replication task. Deploy a replication instance, and configure a change data capture (CDC) task to stream database changes to Amazon S3 as the target. Configure S3 Lifecycle policies to delete the snapshots after 2 years.

AWS DMS focuses more on replication and change data capture rather than backup and restore. It does not address the requirement for daily backups and retention of consistent backups.

Overall Explanation:

The most appropriate solution is to create a backup vault in AWS Backup. This allows for centralized management, a daily schedule, and a 2-year retention policy. AWS Backup is designed for consistent and restorable backups. The other options don't provide the same level of managed backup capabilities.

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Question 4

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Security



A company needs to review its AWS Cloud deployment to ensure that its Amazon S3 buckets do not have unauthorized configuration changes. What should a solutions architect do to accomplish this goal?

- ☒ Turn on AWS Config with the appropriate rules.

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. You can use AWS Config to monitor and record changes to the configuration of your Amazon S3 buckets. By turning on AWS Config and enabling the appropriate rules, you can ensure that your S3 buckets do not have unauthorized configuration changes.

- ☐ Turn on AWS Trusted Advisor with the appropriate checks.

AWS Trusted Advisor provides recommendations to optimize your AWS deployments, improve resilience, and address security gaps, but it doesn't actively prevent or alert on unauthorized configuration changes like AWS Config does.

- ☐ Turn on Amazon Inspector with the appropriate assessment template.

Amazon Inspector is a vulnerability management service that continually scans your AWS workloads for software vulnerabilities and unintended network exposure. It doesn't directly monitor or prevent configuration changes to S3 buckets.

- ☐ Turn on Amazon S3 server access logging. Configure Amazon EventBridge (Amazon CloudWatch Events).

While S3 server access logging and EventBridge can help monitor access and events, they don't provide a mechanism to evaluate and enforce configuration rules to prevent unauthorized changes in the same way as AWS Config.

Overall Explanation:

AWS Config allows you to continuously assess, audit, and evaluate the configurations of your AWS resources, including S3 buckets. You can define rules to monitor and record configuration changes, ensuring that no unauthorized modifications occur. Trusted Advisor provides recommendations, Inspector focuses on vulnerabilities, and S3 logging with EventBridge helps track access, but AWS Config provides configuration management and change prevention.

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Question 5

Not Attempted

Storage



A company's compliance team needs to move its file shares to AWS. The shares run on a Windows Server SMB file share. A self-managed on-premises Active Directory controls access to the files and folders. The company wants to use Amazon FSx for Windows File Server as part of the solution. The company must ensure that the on-premises Active Directory groups restrict access to the FSx for Windows File Server SMB compliance shares, folders, and files after the move to AWS. The company has created an FSx for Windows File Server file system. Which solution will meet these requirements?

- ☐ Create an Active Directory Connector to connect to the Active Directory. Map the Active Directory groups to IAM groups to restrict access.

Creating an Active Directory Connector and mapping groups to IAM groups is more relevant for AWS Directory Service and not for integrating with an existing on-premises Active Directory. Also, Amazon FSx does not support Active Directory Connector.

- ☐ Assign a tag with a Restrict tag key and a Compliance tag value. Map the Active Directory groups to IAM groups to restrict access.

Using tags is typically not used for access control purposes. Tags are metadata and are not directly involved in user authentication and authorization. Also, IAM groups are not used for FSx permissions.

- ☐ Create an IAM service-linked role that is linked directly to FSx for Windows File Server to restrict access.

Creating an IAM service-linked role directly linked to FSx for Windows File Server is not the standard approach for integrating with an existing on-premises Active Directory.

- ☒ Join the file system to the Active Directory to restrict access.

Joining the FSx for Windows File Server file system to the on-premises Active Directory will allow the company to use the existing Active Directory groups to restrict access to the file shares, folders, and files after the move to AWS. This option allows the company to continue using their existing access controls and management structure, making the transition to AWS more seamless.

Overall Explanation:

The best approach is to join the FSx for Windows File Server file system to the on-premises Active Directory. This enables the use of existing Active Directory groups for access control on the file shares, folders, and files after the migration to AWS. This allows the on-premises Active Directory to manage permissions to the FSx file shares, meeting the key requirement to use existing AD groups to control access after migrating to AWS. Other options are not applicable to FSx for Windows File Server.

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Question 6

Not Attempted

[Compute](#)

A company recently deployed its retail website to a global audience. The website runs on multiple Amazon EC2 instances behind an Elastic Load Balancer. The instances are in an Auto Scaling group across multiple Availability Zones. The company wants to provide customers with different versions of content based on the devices they use to access the website. Which combination of actions should a solutions architect take to meet these requirements? (Choose two.)

- ☒ **Configure Amazon CloudFront to cache multiple versions of the content.**
CloudFront can cache different versions of content based on request headers, allowing for device-specific content delivery.
- ☐ **Configure a host header in a Network Load Balancer to forward traffic to different instances.**
Network Load Balancers do not support host-based routing.
- ☒ **Configure a Lambda@Edge function to send specific objects to users based on the User-Agent header.**
Lambda@Edge can inspect the User-Agent header to determine the device type and serve the appropriate content.

- ⊗ Configure AWS Global Accelerator. Forward requests to a Network Load Balancer (NLB). Configure the NLB to set up host-based routing to different EC2 instances.
Network Load Balancers do not support host-based routing.
- ⊗ Configure AWS Global Accelerator. Forward requests to a Network Load Balancer (NLB). Configure the NLB to set up path-based routing to different EC2 instances.
Network Load Balancers do not support path-based routing. NLB just works at layer 4.

Overall Explanation:

The best approach is to use CloudFront to cache multiple versions of the content and use Lambda@Edge to detect the User-Agent header and serve the appropriate version. NLB does not support host or path based routing, making options B, D, and E incorrect. Therefore, options A and C are the correct combination.

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Question 7

Not Attempted

Networking



A company plans to use Amazon ElastiCache for its multi-tier web application. A solutions architect creates a Cache VPC for the ElastiCache cluster and an App VPC for the application's Amazon EC2 instances. Both VPCs are in the us-east-1 Region. The solutions architect must implement a solution to provide the application's EC2 instances with access to the ElastiCache cluster. Which solution will meet these requirements MOST cost-effectively?

- ✔ Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.
Creating a peering connection allows direct and efficient communication. It is the most cost-effective as it avoids the complexity and costs associated with additional resources like a Transit VPC. Also, a more restrictive security group rule can be configured.
- ⊗ Create a Transit VPC. Update the VPC route tables in the Cache VPC and the App VPC to route traffic through the Transit VPC. Configure an inbound rule for the ElastiCache cluster's security group to allow inbound connection from the application's security group.
Creating a Transit VPC adds unnecessary complexity and costs for this scenario.

- ⊗ Create a peering connection between the VPCs. Add a route table entry for the peering connection in both VPCs. Configure an inbound rule for the peering connection's security group to allow inbound connection from the application's security group.
Configuring an inbound rule for the peering connection's security group is not necessary. The ElastiCache cluster's security group should be used.
- ⊗ Create a Transit VPC. Update the VPC route tables in the Cache VPC and the App VPC to route traffic through the Transit VPC. Configure an inbound rule for the Transit VPC's security group to allow inbound connection from the application's security group.
Creating a Transit VPC and configuring its security group adds unnecessary complexity and cost.

Overall Explanation:

VPC peering is the most cost-effective solution. It enables private IP connectivity and allows direct communication between the App VPC and Cache VPC by adding route table entries and configuring the ElastiCache security group. Transit VPCs are more complex and expensive. Configuring the security group for the peering connection itself is not required, only the ElastiCache cluster's security group needs modification.

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Question 8

Not Attempted

[Compute](#)

A company is developing an application composed of several microservices and plans to deploy it on AWS using container technologies. The company's primary goal is to minimize ongoing maintenance and scaling efforts, and they are unwilling to manage additional infrastructure. Which combination of actions should a solutions architect implement to fulfill these requirements? (Choose two.)

- ✔ Deploy an Amazon Elastic Container Service (Amazon ECS) cluster.
Deploying an ECS cluster provides the environment for running containerized applications.
- ⊗ Deploy the Kubernetes control plane on Amazon EC2 instances that span multiple Availability Zones.
Managing the Kubernetes control plane on EC2 instances requires significant infrastructure management, which contradicts the requirement of minimizing effort.

- ⊗ Deploy an Amazon Elastic Container Service (Amazon ECS) service with an Amazon EC2 launch type. Specify a desired task number level of greater than or equal to 2.
Using the EC2 launch type necessitates managing the underlying EC2 instances, increasing maintenance overhead.
- ✔ Deploy an Amazon Elastic Container Service (Amazon ECS) service with a Fargate launch type. Specify a desired task number level of greater than or equal to 2.
Utilizing the Fargate launch type eliminates the need to manage the underlying infrastructure, allowing ECS to handle provisioning and scaling automatically.
- ⊗ Deploy Kubernetes worker nodes on Amazon EC2 instances that span multiple Availability Zones. Create a deployment that specifies two or more replicas for each microservice.
Similar to option B, this involves managing EC2 instances for the worker nodes, adding to the operational burden.

Overall Explanation:

The ideal solution involves using Amazon ECS with the Fargate launch type. ECS provides a container orchestration service, while Fargate allows running containers without managing servers or clusters of Amazon EC2 instances. Deploying an ECS cluster and using the Fargate launch type with a desired task count of at least 2 ensures high availability and minimal management overhead, perfectly aligning with the company's requirements. Options B, C and E all involve managing EC2 instances, contradicting the requirement to minimize infrastructure management.

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Question 9

Not Attempted

Networking



A company hosts its web application on 10 Amazon EC2 instances with traffic managed by Amazon Route 53. Users are reporting intermittent timeout errors when accessing the application. The networking team has determined that some DNS queries are resolving to the IP addresses of unhealthy instances, which is causing these timeouts. What is the MOST suitable solution a solutions architect should implement to resolve the timeout errors and ensure high availability?

- ⊗ Create a Route 53 simple routing policy record for each EC2 instance and associate a health check with each record.

While this configures health checks, it does not provide automatic load balancing or traffic distribution to healthy instances, potentially still resulting in timeouts due to DNS caching.

- ⊗ Create a Route 53 failover routing policy record for each EC2 instance and associate a health check with each record.

Failover routing is typically used for active-passive setups and might not efficiently utilize all healthy instances for load balancing, and DNS caching can still lead to timeouts.

- ⊗ Create an Amazon CloudFront distribution with the EC2 instances as origins and associate a health check with the EC2 instances.

CloudFront is primarily a CDN and focuses on caching content closer to users, and while it offers some health check capabilities, it doesn't directly address load balancing across healthy instances as effectively as an ALB.

- ✔ Create an Application Load Balancer (ALB) with a health check in front of the EC2 instances. Route traffic to the ALB from Route 53.

The ALB performs health checks on the EC2 instances and routes traffic only to healthy instances, thus preventing timeouts. Route 53 directs traffic to the ALB, which then manages the load balancing and health checks. This is a standard and robust architecture for public-facing web applications.

Overall Explanation:

The most effective solution is to use an Application Load Balancer (ALB) with health checks. The ALB will actively monitor the health of each EC2 instance and only route traffic to those that are healthy. By routing traffic from Route 53 to the ALB, you ensure that users are only directed to available and responsive instances, thus eliminating the timeout errors. The ALB handles the distribution of traffic across the healthy instances, providing enhanced availability and scalability. Options A and B don't provide automatic load balancing and option C is a CDN, which is not designed for health based routing to backend instances.

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Question 10

Not Attempted

Networking



A solutions architect is designing a highly available application with web, application, and database tiers. They need to deliver HTTPS content with minimal latency and ensure high security by minimizing exposure to the public internet. Which solution best meets these requirements?

- ⊗ Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in public subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.

This option uses public subnets for the EC2 instances, increasing the attack surface and security risk.

- ⊗ Configure a public Application Load Balancer with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the EC2 instances as the origin.

This option attempts to use EC2 instances in private subnets as the origin for CloudFront which is incorrect. CloudFront should be using an ALB.

- ✔ Configure a public Application Load Balancer (ALB) with multiple redundant Amazon EC2 instances in private subnets. Configure Amazon CloudFront to deliver HTTPS content using the public ALB as the origin.

This option is the most secure because the EC2 instances are in private subnets, and CloudFront caches content from the ALB for faster delivery. It provides edge-based content delivery, high availability, and secure access by limiting direct access to the web servers.

- ⊗ Configure a public Application Load Balancer with multiple redundant Amazon EC2 instances in public subnets. Configure Amazon CloudFront to deliver HTTPS content using the EC2 instances as the origin.

This option places EC2 instances in public subnets, exposing them to the internet directly, which presents a higher security risk and does not leverage an ALB as an origin.

Overall Explanation:

The best solution is to configure a public ALB with EC2 instances in private subnets. CloudFront is then configured to use the public ALB as its origin. This setup provides high availability, secure access through private subnets, and optimized HTTPS content delivery using CloudFront. The EC2 instances being in private subnets reduces the attack surface, making it the most secure option. CloudFront delivers content close to the edge, and the ALB ensures traffic distribution to healthy instances.

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Question 11

Not Attempted

Networking



A company hosts a globally distributed, latency-sensitive gaming platform on AWS. The application is deployed across multiple AWS Regions and runs on EC2 instances within Auto Scaling groups behind Application Load Balancers (ALBs). To ensure optimal user experience and fair gameplay, the solutions architect needs to implement a mechanism that monitors application health and redirects traffic to the closest healthy endpoint with minimal latency.

Which solution best meets these requirements?

- ☒ Configure an accelerator in AWS Global Accelerator. Add a listener for the port that the application listens on, and attach it to a Regional endpoint in each Region. Add the ALB as the endpoint.

AWS Global Accelerator directs traffic to the optimal healthy endpoint based on health checks and geographic proximity. By adding the ALBs in each region as endpoints to the Global Accelerator, traffic will be routed to the closest and healthiest available endpoint, minimizing latency.

- ☐ Create an Amazon CloudFront distribution and specify the ALB as the origin server. Configure the cache behavior to use origin cache headers. Use AWS Lambda functions to optimize the traffic.

CloudFront is primarily a content delivery network (CDN) and focuses on caching static content closer to users. While it can improve latency for cached content, it's not the best solution for dynamically routing traffic based on health and proximity for real-time gaming applications.

- ☐ Create an Amazon CloudFront distribution and specify Amazon S3 as the origin server. Configure the cache behavior to use origin cache headers. Use AWS Lambda functions to optimize the traffic.

Using S3 as the origin with CloudFront is suitable for serving static assets but not for routing dynamic game traffic based on health and location.

- ☐ Configure an Amazon DynamoDB database to serve as the data store for the application. Create a DynamoDB Accelerator (DAX) cluster to act as the in-memory cache for DynamoDB hosting the application data.

DynamoDB and DAX are focused on improving data access latency. While important for the game backend, it does not address the requirement to route user traffic to healthy endpoints with minimal latency.

Overall Explanation:

The best solution is to use AWS Global Accelerator. It monitors the health of application endpoints in different AWS Regions and intelligently routes traffic to the closest and healthiest endpoint, minimizing latency for users. The other options, such as CloudFront and DynamoDB Accelerator (DAX), do not provide the required functionality for dynamic traffic routing based on health and proximity. Global Accelerator can be used for non-http cases such as UDP, tcp, gaming, or VOIP.

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Question 12

Not Attempted

Database



A company has one million users that use its mobile app. The company must analyze the data usage in near-real time. The company also must encrypt the data in near-real time and must store the data in a centralized location in Apache Parquet format for further processing. Which solution will meet these requirements with the LEAST operational overhead?

- ☐ Create an Amazon Kinesis data stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data. Invoke an AWS Lambda function to send the data to the Kinesis Data Analytics application.

This option requires invoking a Lambda function, introducing additional operational overhead and complexity.

- ☐ Create an Amazon Kinesis data stream to store the data in Amazon S3. Create an Amazon EMR cluster to analyze the data. Invoke an AWS Lambda function to send the data to the EMR cluster.

While EMR is powerful, it requires more operational management compared to Kinesis Data Analytics and involves a Lambda function.

- ☐ Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon EMR cluster to analyze the data.

This introduces unnecessary complexity by involving EMR for data analysis when Kinesis Data Analytics can perform the analysis in a more streamlined manner.

- ☒ Create an Amazon Kinesis Data Firehose delivery stream to store the data in Amazon S3. Create an Amazon Kinesis Data Analytics application to analyze the data.

This solution leverages Kinesis Data Firehose for data ingestion and storage in S3 (Parquet format) and Kinesis Data Analytics for near-real-time analysis, providing a low operational overhead.

Overall Explanation:

Kinesis Data Firehose automatically handles data collection, transformation, encryption, and storage in near-real time, directly to S3 in Apache Parquet format. Kinesis Data Analytics then provides near-real-time analysis without needing infrastructure management or Lambda functions, resulting in the least operational overhead. Kinesis Data Streams cannot directly send data to S3, and using EMR increases operational overhead compared to Kinesis Data Analytics.

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Question 13

Not Attempted

Database



A gaming company's web application, hosted on EC2 instances behind an Application Load Balancer, displays scores. The application uses an Amazon RDS for MySQL database. Users are experiencing delays and interruptions due to poor database read performance. The company aims to enhance the user experience with minimal architectural changes. What should a solutions architect do to meet these requirements?

☒ Use Amazon ElastiCache in front of the database.

ElastiCache can improve read performance by caching frequently accessed data, reducing the load on the database and improving response times. This approach minimizes changes to the application architecture compared to database migrations or major code refactoring.

☐ Use RDS Proxy between the application and the database.

RDS Proxy is designed to manage database connections, pooling, and failover, improving database scalability and performance. It helps reduce the number of open connections to the database, which can be a common source of performance issues. However, the problem is read performance and not the number of connections.

☐ Migrate the application from EC2 instances to AWS Lambda.

Migrating to Lambda would involve significant changes to the application's architecture and codebase, which contradicts the requirement of minimizing changes.

☐ Migrate the database from Amazon RDS for MySQL to Amazon DynamoDB.

Migrating to DynamoDB, a NoSQL database, would require substantial changes to the application's data model and query patterns, violating the requirement of minimizing architectural changes.

Overall Explanation:

Using Amazon ElastiCache (Option A) is the most suitable solution. It addresses the database read performance issue by caching frequently accessed data, improving response times. It also allows the company to improve performance without requiring significant changes to the application architecture. Introducing caching requires some code changes, but these are less impactful compared to migrating the database or changing the application's underlying architecture. RDS Proxy is more focused on connection management. Options C and D require major architectural changes.

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Question 14

Not Attempted

Database



An e-commerce company's web application, backed by an Amazon RDS database, is experiencing performance degradation due to an increased number of read-only SQL queries run by business analysts. A solutions architect needs to address this issue with minimal disruption to the existing web application. Which of the following solutions should the architect recommend?

- ☐ Export the data to Amazon DynamoDB and have the business analysts run their queries.
DynamoDB is a NoSQL database requiring significant changes to the application's data model and query patterns, making it a disruptive solution.
- ☐ Load the data into Amazon ElastiCache and have the business analysts run their queries.
ElastiCache is primarily a caching layer, not suitable for running complex analytical queries directly. Implementing this would require code changes for caching and retrieval.
- ☒ Create a read replica of the primary database and have the business analysts run their queries.
Creating a read replica allows offloading read-only queries without modifying the core application. The business analysts can query the replica, minimizing impact on the primary database's performance.
- ☐ Copy the data into an Amazon Redshift cluster and have the business analysts run their queries.

Redshift is a data warehouse solution requiring significant data migration and query adaptation, representing a major change to the application's architecture.

Overall Explanation:

The most suitable solution is to create a read replica of the primary RDS database. This approach allows business analysts to run their read-only queries on the replica, thus offloading the primary database and resolving the performance bottleneck with minimal disruption to the existing web application. Options A, B, and D all involve more significant changes to the application architecture and data management.

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Question 15

Not Attempted

Monitoring



A company is launching a new application and wants to display application metrics on an Amazon CloudWatch dashboard. The company's product manager needs to access this dashboard periodically but does not have an AWS account. A solutions architect must provide access to the product manager while adhering to the principle of least privilege. Which solution will BEST meet these requirements?

- ☒ Share the dashboard from the CloudWatch console. Enter the product manager's email address, and complete the sharing steps. Provide a shareable link for the dashboard to the product manager.

Sharing the dashboard via email creates an IAM role behind the scenes with limited permissions, adhering to the principle of least privilege. It also doesn't require the product manager to have an AWS account.

- ☐ Create an IAM user specifically for the product manager. Attach the CloudWatchReadOnlyAccess AWS managed policy to the user. Share the new login credentials with the product manager. Share the browser URL of the correct dashboard with the product manager.

While this provides access, CloudWatchReadOnlyAccess grants broader permissions than necessary, violating the principle of least privilege. Also, it requires an AWS account for the product manager, which they do not have.

- ☐ Create an IAM user for the company's employees. Attach the ViewOnlyAccess AWS managed policy to the IAM user. Share the new login credentials with the product

manager. Ask the product manager to navigate to the CloudWatch console and locate the dashboard by name in the Dashboards section.

This option provides access to other dashboards in the account which violates the principle of least privilege, also requires an AWS account for the product manager, which they do not have.

- ⊗ Deploy a bastion server in a public subnet. When the product manager requires access to the dashboard, start the server and share the RDP credentials. On the bastion server, ensure that the browser is configured to open the dashboard URL with cached AWS credentials that have appropriate permissions to view the dashboard.

This is overly complex and unnecessary. It involves creating and managing infrastructure (bastion server) and does not follow the principle of least privilege. It also requires setting up RDP and cached credentials, adding significant overhead.

Overall Explanation:

The best solution is to share the CloudWatch dashboard by providing the product manager's email address. This method allows access to the specific dashboard without requiring an AWS account and grants minimal necessary permissions through an IAM role created behind the scenes. This adheres to the principle of least privilege by granting only the necessary permissions to view the specific dashboard.

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Question 16

Not Attempted

Security



A company is using a centralized AWS account to store log data in various Amazon S3 buckets. A solutions architect needs to ensure that the data is encrypted at rest **before** the data is uploaded to the S3 buckets. The data also must be encrypted in transit. Which solution meets these requirements?

- ✔ Use client-side encryption to encrypt the data that is being uploaded to the S3 buckets.
Client-side encryption ensures that the data is encrypted before it is sent to S3, meeting the requirement of encryption at rest before upload. Using HTTPS also encrypts the data in transit.
- ⊗ Use server-side encryption to encrypt the data that is being uploaded to the S3 buckets.

Server-side encryption encrypts the data after it has been uploaded to S3. This does not meet the requirement of encrypting the data **before** upload.

- ⊗ Create bucket policies that require the use of server-side encryption with S3 managed encryption keys (SSE-S3) for S3 uploads.

Requiring server-side encryption via bucket policies encrypts data at rest, but only after it's been uploaded. This does not satisfy the 'before upload' requirement.

- ⊗ Enable the security option to encrypt the S3 buckets through the use of a default AWS Key Management Service (AWS KMS) key.

Enabling default KMS encryption encrypts data at rest, but only after it's been uploaded. This does not satisfy the 'before upload' requirement.

Overall Explanation:

The key requirement is that the data must be encrypted **before** it is uploaded to S3.

Client-side encryption allows you to encrypt the data locally before transmitting it to S3.

Server-side encryption options encrypt the data after it reaches S3. HTTPS satisfies the in-transit encryption requirement.

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Question 17

Not Attempted

[Compute](#)

A solutions architect observes that a nightly batch processing job automatically scales up for 1 hour before the desired Amazon EC2 capacity is reached. The peak capacity is the same every night, and the batch jobs always start at 1 AM. The solutions architect needs to find a cost-effective solution that will allow the desired EC2 capacity to be reached quickly and allow the Auto Scaling group to scale down after the batch jobs are complete. What should the solutions architect do to meet these requirements?

- ⊗ Increase the minimum capacity for the Auto Scaling group.

Increasing the minimum capacity would ensure instances are always running, which isn't cost-effective and doesn't guarantee reaching desired capacity immediately at 1 AM.

- ⊗ Increase the maximum capacity for the Auto Scaling group.

Increasing the maximum capacity doesn't solve the problem of reaching the desired capacity quickly at the specific time.

- ✔ Configure scheduled scaling to scale up to the desired compute level.

Scheduled scaling allows setting the Auto Scaling group to automatically scale up to the desired compute level at a specific time (1 AM), ensuring desired capacity is reached immediately. It can also scale down after the job completes, optimizing costs.

⊗ Change the scaling policy to add more EC2 instances during each scaling operation.

While this might help scale up faster, it doesn't guarantee reaching the exact desired capacity at 1 AM and might lead to over-provisioning or needing more fine-tuning. Scheduled scaling is more precise for predictable workloads.

Overall Explanation:

Configuring scheduled scaling actions allows the Auto Scaling group to scale up to the desired capacity at a scheduled time (1 AM in this case) when the batch jobs start. This ensures the desired compute capacity is reached immediately. The Auto Scaling group can then scale down based on metrics after the batch jobs complete, making it the most cost-effective and efficient solution for this scenario. Using scheduled scaling caters for predictable scaling needs.

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Question 18

Not Attempted

Networking



A company hosts a dynamic website on Amazon EC2 instances behind an Application Load Balancer (ALB) in the us-west-1 Region. The website serves a global audience and supports multiple languages. Users in distant locations are experiencing high request latency. The company wants to improve response times for all users without replicating the entire architecture across multiple AWS Regions. What should a solutions architect do to minimize latency for global users most cost-effectively?

⊗ Replace the existing architecture with a static website served from an Amazon S3 bucket. Configure an Amazon CloudFront distribution with the S3 bucket as the origin. Configure cache behavior settings to cache based on the Accept-Language request header.

This approach is not ideal because the website is dynamic, and S3 is better suited for static content. Also, completely replacing the existing architecture is unnecessary.

✔ Configure an Amazon CloudFront distribution with the ALB as the origin. Set the cache behavior settings to cache based on the Accept-Language request header.

This is the most suitable solution. CloudFront can cache content closer to users globally, reducing latency. Caching based on the Accept-Language header ensures users receive the correct language version of the website.

- ⊗ Create an Amazon API Gateway API integrated with the ALB using HTTP integration. Configure the API Gateway stage to enable API caching based on the Accept-Language request header.

While API Gateway can cache, it adds unnecessary complexity and cost compared to using CloudFront directly with the ALB.

- ⊗ Launch an EC2 instance in each additional Region and configure NGINX to act as a cache server for that Region. Place all EC2 instances and the ALB behind an Amazon Route 53 record set with a geolocation routing policy.

This is a complex and costly solution involving managing multiple EC2 instances and configuring NGINX in each region. CloudFront provides a managed caching service that is more efficient and scalable.

Overall Explanation:

The best solution is to use Amazon CloudFront with the ALB as the origin. By configuring CloudFront to cache based on the Accept-Language header, the appropriate version of the website is served to users based on their language preferences, reducing latency and improving performance globally without the need to replicate infrastructure across multiple regions.

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Question 19

Not Attempted

Database



A rapidly growing e-commerce company currently operates its workloads in a single AWS Region. To enhance business continuity, a solutions architect needs to design a disaster recovery (DR) strategy involving a secondary AWS Region. The key requirements are maintaining a near real-time database copy in the DR Region with minimal latency and running the remaining infrastructure at reduced capacity, capable of scaling up on demand in the event of a disaster. Which solution provides the LOWEST Recovery Time Objective (RTO)?

- ⊗ Use an Amazon Aurora global database with a pilot light deployment.

A pilot light deployment, while cost-effective, requires manual intervention to fully activate the DR Region, resulting in a higher RTO. The infrastructure needs to be 'turned on' and scaled up, delaying recovery.

- ✔ Use an Amazon Aurora global database with a warm standby deployment.

A warm standby deployment maintains a scaled-down, but fully functional, environment in the DR Region, including an up-to-date database copy. This allows for faster recovery with minimal latency, achieving the lowest RTO.

- ✘ Use an Amazon RDS Multi-AZ DB instance with a pilot light deployment.

Amazon RDS Multi-AZ provides high availability *within* a single Region, not disaster recovery *across* Regions. Also, pilot light configuration will increase the recovery time.

- ✘ Use an Amazon RDS Multi-AZ DB instance with a warm standby deployment.

Amazon RDS Multi-AZ provides high availability *within* a single Region, not disaster recovery *across* Regions, therefore it does not meet the requirement of a DR strategy in a different AWS Region.

Overall Explanation:

The requirement for a disaster recovery strategy in a different AWS Region with the LOWEST RTO dictates the use of Amazon Aurora Global Database with a Warm Standby deployment. This architecture keeps the database updated in the DR region with minimal latency and allows for the remaining infrastructure to run at reduced capacity but ready to scale up. Pilot Light deployment requires manual intervention, which increases RTO, and Multi-AZ configuration only works within the same region.

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Question 20

Not Attempted

Management & Governance



A company runs an application on Amazon EC2 instances. The company needs to implement a disaster recovery (DR) solution for the application with a recovery time objective (RTO) of less than 4 hours. The DR solution must also minimize AWS resource utilization during normal operations. Which solution provides the MOST operationally efficient way to meet these requirements?

- ⊗ Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS Lambda and custom scripts.

While this option uses AMIs and a secondary region, using Lambda and custom scripts for infrastructure deployment is less operationally efficient than using a dedicated infrastructure-as-code tool like CloudFormation. Custom scripts require more manual maintenance and are prone to errors.

- ✔ Create Amazon Machine Images (AMIs) to back up the EC2 instances. Copy the AMIs to a secondary AWS Region. Automate infrastructure deployment in the secondary Region by using AWS CloudFormation.

This option utilizes AMIs for backups and CloudFormation for automated infrastructure deployment in the secondary region. CloudFormation allows for a declarative and repeatable approach to infrastructure provisioning, minimizing manual intervention and operational overhead. This provides a good balance of RTO and resource utilization during normal operations.

- ⊗ Launch EC2 instances in a secondary AWS Region. Keep the EC2 instances in the secondary Region active at all times.

This solution provides a fast RTO but is not operationally efficient. Keeping EC2 instances active in a secondary region at all times significantly increases costs and resource consumption during normal operations, violating the requirement to use the fewest possible AWS resources.

- ⊗ Launch EC2 instances in a secondary Availability Zone. Keep the EC2 instances in the secondary Availability Zone active at all times.

While providing high availability, this option only protects against failures within a single AWS Region and does not constitute a disaster recovery solution that utilizes a separate region. Keeping EC2 instances active also increases costs and resource consumption during normal operations.

Overall Explanation:

Option B provides the most operationally efficient DR solution. It leverages AMIs for backups and CloudFormation for infrastructure automation in a secondary AWS Region. CloudFormation's infrastructure-as-code approach allows for repeatable deployments, reducing the burden of manual configuration and management. Options C and D fail to meet the requirement of using the fewest possible AWS resources during normal operations, and option A is less efficient due to the use of Lambda and custom scripts compared to the purpose built CloudFormation.

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Question 21

Not Attempted

Compute



A company hosts an internal browser-based application on Amazon EC2 instances behind an Application Load Balancer. The EC2 instances reside in an Auto Scaling group across multiple Availability Zones. During work hours, the Auto Scaling group scales up to 20 instances, but scales down to 2 instances overnight. Staff report that the application is very slow when the day begins, but performance improves by mid-morning. Which scaling change will best address the staff complaints while minimizing costs?

- ☐ Implement a scheduled action that sets the desired capacity to 20 shortly before the office opens.

Setting the desired capacity to 20 may keep it at 20 for the entire day, even when the load is reduced, leading to unnecessary costs.

- ☐ Implement a step scaling action triggered at a lower CPU threshold, and decrease the cooldown period.

Step scaling with small steps might scale too slowly. Larger steps might not be cost-efficient during the day when demand fluctuates.

- ☒ Implement a target tracking action triggered at a lower CPU threshold, and decrease the cooldown period.

Target tracking scaling allows the Auto Scaling group to adjust capacity in response to demand while maintaining a specific metric (e.g., CPU utilization) at the target value. A lower threshold triggers scaling earlier, and a decreased cooldown period allows for faster scaling adjustments.

- ☐ Implement a scheduled action that sets the minimum and maximum capacity to 20 shortly before the office opens.

Setting both the minimum and maximum capacity to 20 would ensure 20 instances are always running, regardless of actual demand, leading to higher costs.

Overall Explanation:

Target tracking offers the most dynamic and cost-effective solution. By triggering at a lower CPU threshold, the Auto Scaling group proactively adds instances as workload increases, addressing the morning slowness. Decreasing the cooldown period allows for quicker scaling adjustments. This approach is preferable to scheduled actions, which don't account for actual demand, and step scaling, which can be inefficient. The selected

approach also scales down resources automatically during off-peak hours, keeping costs to a minimum.

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Question 22

Not Attempted

[Compute](#)

A company's multi-tier application, deployed on Amazon EC2 instances within an Auto Scaling group, uses an Amazon RDS for Oracle instance with Oracle-specific PL/SQL functions as its data layer. The application experiences steadily increasing traffic, causing EC2 instance overload and RDS storage exhaustion. The Auto Scaling group is configured only with a minimum healthy instance count and lacks scaling metrics. The company anticipates continued, unpredictable traffic growth before stabilization. What two actions should a solutions architect take to ensure the system automatically scales to handle the increased traffic?

- ☒ **Configure storage Auto Scaling on the RDS for Oracle instance.**
Enabling storage Auto Scaling allows the RDS instance to automatically increase its allocated storage when free storage space gets low, preventing the database from running out of capacity.
- ☐ **Migrate the database to Amazon Aurora to use Auto Scaling storage.**
Migrating to Aurora is not a viable option because the application utilizes Oracle-specific PL/SQL functions, which are not directly compatible with Aurora.
- ☐ **Configure an alarm on the RDS for Oracle instance for low free storage space.**
Configuring an alarm provides monitoring but doesn't automatically scale resources. It requires manual intervention to address storage issues.
- ☒ **Configure the Auto Scaling group to use the average CPU utilization as the scaling metric.**
Using average CPU utilization as a scaling metric allows the Auto Scaling group to dynamically adjust the number of EC2 instances based on application demand, handling increased traffic and preventing instance overload.
- ☐ **Configure the Auto Scaling group to use the average free memory as the scaling metric.**
While memory is important, CPU utilization is a more common and direct indicator of application load for scaling EC2 instances.

Overall Explanation:

To automatically scale for increased traffic, the solutions architect should configure storage Auto Scaling on the RDS for Oracle instance to dynamically manage storage capacity and configure the Auto Scaling group to use average CPU utilization as the scaling metric, allowing it to launch additional instances based on increased demand. This addresses both the database storage issue and the EC2 instance overload.

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Question 23

Not Attempted

Storage



A company provides an online service for posting video content and transcoding it for use by any mobile platform. The application architecture uses Amazon Elastic File System (Amazon EFS) Standard to collect and store the videos so that multiple Amazon EC2 Linux instances can access the video content for processing. As the popularity of the service has grown, the storage costs have become too expensive. Which storage solution is MOST cost-effective?

- ☐ Use AWS Storage Gateway for files to store and process the video content.
AWS Storage Gateway is not typically used for storing content, but rather for transferring data to the cloud. It can be used to provide a file interface to S3, but by itself it does not offer cost savings in this scenario.
- ☐ Use AWS Storage Gateway for volumes to store and process the video content.
Similar to the file gateway, the volume gateway is not a storage solution on its own but rather a way to interface with cloud storage. It wouldn't address the cost issue with EFS.
- ☐ Use Amazon EFS for storing the video content. Once processing is complete, transfer the files to Amazon Elastic Block Store (Amazon EBS).
Using EFS for initial storage and then moving to EBS would likely be more expensive than the current EFS setup, as EBS is generally more expensive than S3 for storage at rest.
- ☒ Use Amazon S3 for storing the video content. Move the files temporarily over to an Amazon Elastic Block Store (Amazon EBS) volume attached to the server for processing.
Amazon S3 provides low-cost object storage suitable for storing large amounts of unstructured data like videos. Moving the files to EBS temporarily for processing allows leveraging EBS's high performance only when needed, minimizing storage costs.

Overall Explanation:

Amazon S3 is the most cost-effective option for storing large video files. By moving the files to EBS only for the duration of the processing, the overall costs can be minimized. Options A and B involve Storage Gateway, which is typically used for hybrid cloud scenarios and doesn't inherently lower storage costs. Option C maintains the use of EFS which is stated to be too expensive already, and adds EBS, resulting in increased costs. Therefore, option D is the most suitable solution.

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Question 24

Not Attempted

Database



A company wants to create an application to store employee data in a hierarchical structured relationship. The company needs a minimum-latency response to high-traffic queries for the employee data and must protect any sensitive data. The company also needs to receive monthly email messages if any financial information is present in the employee data. Which combination of steps should a solutions architect take to meet these requirements?

- ☐ Use Amazon Redshift to store the employee data in hierarchies. Unload the data to Amazon S3 every month.

Amazon Redshift is not optimal for hierarchical data and frequent, low-latency queries.

- ☒ Use Amazon DynamoDB to store the employee data in hierarchies. Export the data to Amazon S3 every month.

DynamoDB is suitable for storing hierarchical data and provides low-latency, high-performance storage for high-traffic queries. Exporting to S3 provides an audit trail and backup.

- ☐ Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly events to AWS Lambda.

While Macie is helpful for sensitive data discovery, Lambda alone won't send email notifications.

- ☐ Use Amazon Athena to analyze the employee data in Amazon S3. Integrate Athena with Amazon QuickSight to publish analysis dashboards and share the dashboards with users.

Athena and QuickSight are suitable for analytics, but don't address the low-latency requirement or email notifications.

- ✓ Configure Amazon Macie for the AWS account. Integrate Macie with Amazon EventBridge to send monthly notifications through an Amazon Simple Notification Service (Amazon SNS) subscription.

Macie detects sensitive data. Integrating with EventBridge and SNS allows for sending email notifications when sensitive data is found.

Overall Explanation:

Storing employee data in DynamoDB allows for low-latency retrieval of data in a hierarchical structure. DynamoDB handles high-traffic queries. Exporting the data to S3 monthly enables audit trails. Amazon Macie can scan the data for sensitive information, like financial data. Integrating Macie with EventBridge and SNS enables monthly email notifications when sensitive data is detected.

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Question 25

Not Attempted

Database



A company has an application that is backed by an Amazon DynamoDB table. The company's compliance requirements specify that database backups must be taken every month, must be available for 6 months in warm storage, and must be retained for 7 years.

- ✓ Create an AWS Backup plan to back up the DynamoDB table on the first day of each month. Specify a lifecycle policy that transitions the backup to cold storage after 6 months. Set the retention period for each backup to 7 years.

This solution utilizes AWS Backup, which is designed for managing backups. It allows for scheduled backups, lifecycle policies to transition to cold storage, and setting retention periods to meet compliance requirements efficiently.

- ✗ Create a DynamoDB on-demand backup of the DynamoDB table on the first day of each month. Transition the backup to Amazon S3 Glacier Flexible Retrieval after 6 months. Create an S3 Lifecycle policy to delete backups that are older than 7 years.
DynamoDB doesn't natively integrate with S3 Glacier Flexible Retrieval for backup management. Using S3 introduces unnecessary complexity and is not the standard practice for DynamoDB backup lifecycle management.
- ✗ Use the AWS SDK to develop a script that creates an on-demand backup of the DynamoDB table. Set up an Amazon EventBridge rule that runs the script on the first

day of each month. Create a second script that will run on the second day of each month to transition DynamoDB backups that are older than 6 months to cold storage and to delete backups that are older than 7 years.

This approach introduces unnecessary complexity through custom scripting. AWS Backup provides a managed service designed for this purpose, simplifying the process and reducing the risk of errors.

- ⊗ Use the AWS CLI to create an on-demand backup of the DynamoDB table. Set up an Amazon EventBridge rule that runs the command on the first day of each month with a cron expression. Specify in the command to transition the backups to cold storage after 6 months and to delete the backups after 7 years.

Similar to Option C, this approach uses custom scripting and adds complexity. AWS Backup offers a more straightforward and managed solution for scheduling backups and managing retention policies.

Overall Explanation:

Option A is the correct answer. AWS Backup is a managed service designed to handle backups, retention, and lifecycle policies for DynamoDB. It allows for scheduled backups, transitions to cold storage, and setting retention periods, meeting all requirements.

Options B, C, and D involve manual or scripted processes that are more complex and less efficient than using the built-in capabilities of AWS Backup. DynamoDB backups do not natively integrate with S3 Glacier. Using custom scripts increases the risk of errors and operational overhead.

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Question 26

Not Attempted

Security



A company is migrating applications to AWS. The applications are deployed in different accounts. The company manages the accounts centrally by using AWS Organizations. The company's security team needs a single sign-on (SSO) solution across all the company's accounts. The company must continue managing the users and groups in its on-premises self-managed Microsoft Active Directory. Which solution will meet these requirements?

- ⊗ Enable AWS Single Sign-On (AWS SSO) from the AWS SSO console. Create a one-way forest trust or a one-way domain trust to connect the company's self-managed Microsoft Active Directory with AWS SSO by using AWS Directory Service for Microsoft Active Directory.

This option may seem plausible, but AWS SSO and applications like Amazon Chime, Amazon Connect, Amazon QuickSight, Amazon WorkDocs, Amazon WorkMail, Amazon WorkSpaces and the AWS Management Console typically require a two-way trust for full functionality when integrating with a self-managed Active Directory.

- ✔ Enable AWS Single Sign-On (AWS SSO) from the AWS SSO console. Create a two-way forest trust to connect the company's self-managed Microsoft Active Directory with AWS SSO by using AWS Directory Service for Microsoft Active Directory.

This is the correct solution. A two-way forest trust allows AWS Managed Microsoft AD to query users and groups in the on-premises self-managed AD, which is necessary for AWS SSO and many AWS Enterprise Applications like Amazon Chime, Amazon Connect and Amazon Workspaces.

- ✘ Use AWS Directory Service. Create a two-way trust relationship with the company's self-managed Microsoft Active Directory.

This option is incomplete. While creating a two-way trust is necessary, this option doesn't mention enabling AWS SSO, which is required for single sign-on across the company's AWS accounts.

- ✘ Deploy an identity provider (IdP) on premises. Enable AWS Single Sign-On (AWS SSO) from the AWS SSO console.

The company already has an on-premises Active Directory and doesn't need to deploy a separate IdP. The focus is on integrating the existing Active Directory with AWS SSO, which makes this option less suitable.

Overall Explanation:

The correct answer is to enable AWS Single Sign-On (AWS SSO) and create a two-way forest trust using AWS Directory Service for Microsoft Active Directory. A two-way trust is required for AWS Enterprise Apps such as Amazon Chime, Amazon Connect, Amazon QuickSight, AWS IAM Identity Center, Amazon WorkDocs, Amazon WorkMail, Amazon WorkSpaces, and the AWS Management Console. AWS Managed Microsoft AD must be able to query the users and groups in your self-managed AD.

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Question 27

Not Attempted

Management & Governance



A company is using Amazon CloudFront with its website. The company has enabled logging on the CloudFront distribution, and logs are saved in one of the company's Amazon S3 buckets.

The company needs to perform advanced analyses on the logs and build visualizations. What should a solutions architect do to meet these requirements?

- ⊗ Use standard SQL queries in Amazon Athena to analyze the CloudFront logs in the S3 bucket. Visualize the results with AWS Glue.

AWS Glue is primarily used for data transformation and ETL (Extract, Transform, Load) processes, not for creating visualizations. While Glue can prepare data for analysis, it doesn't directly offer visualization capabilities like Amazon QuickSight.

- ✓ Use standard SQL queries in Amazon Athena to analyze the CloudFront logs in the S3 bucket. Visualize the results with Amazon QuickSight.

Amazon Athena allows you to run standard SQL queries on data stored in Amazon S3. Amazon QuickSight is a business intelligence (BI) service that enables you to create interactive visualizations and dashboards. This combination is ideal for analyzing CloudFront logs and building visualizations.

- ⊗ Use standard SQL queries in Amazon DynamoDB to analyze the CloudFront logs in the S3 bucket. Visualize the results with AWS Glue.

Amazon DynamoDB is a NoSQL database, not suitable for directly querying CloudFront logs stored in S3 using standard SQL. Additionally, AWS Glue is not a visualization tool.

- ⊗ Use standard SQL queries in Amazon DynamoDB to analyze the CloudFront logs in the S3 bucket. Visualize the results with Amazon QuickSight.

Amazon DynamoDB is not designed for querying data in S3. CloudFront logs are stored in S3, making Athena the correct choice for querying. While QuickSight is a good visualization tool, it needs to be used with a service that can access and query the data in S3.

Overall Explanation:

The optimal solution is to use Amazon Athena to query the CloudFront logs directly from the S3 bucket using standard SQL queries. The results can then be visualized using Amazon QuickSight, which is designed for creating interactive dashboards and visualizations. Athena's ability to query S3 data combined with QuickSight's visualization capabilities provides a cost-effective and efficient way to meet the requirements.

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Question 28

Not Attempted

Database



A company runs a fleet of web servers using an Amazon RDS for PostgreSQL DB instance. After a routine compliance check, the company sets a standard that requires a recovery point objective (RPO) of less than 1 second for all its production databases. Which solution meets these requirements?

- ☒ Enable a Multi-AZ deployment for the DB instance.

Multi-AZ deployments provide synchronous replication to a standby instance in a different Availability Zone. This synchronous replication ensures that data changes are continuously replicated, providing an RPO of less than 1 second. In case of a failure, the standby instance automatically takes over, minimizing data loss.

- ☐ Enable auto scaling for the DB instance in one Availability Zone.

Auto scaling adjusts the resources of a DB instance based on demand but doesn't provide redundancy or minimize data loss in the event of a failure. It does not directly contribute to meeting the RPO requirement.

- ☐ Configure the DB instance in one Availability Zone, and create multiple read replicas in a separate Availability Zone.

Read replicas use asynchronous replication, meaning there's a delay between changes on the primary instance and their replication to the read replicas. This delay makes it unsuitable for meeting an RPO of less than 1 second.

- ☐ Configure the DB instance in one Availability Zone, and configure AWS Database Migration Service (AWS DMS) change data capture (CDC) tasks.

AWS DMS with CDC can replicate data, but it's primarily used for migrating data between databases or for data warehousing purposes. While DMS CDC can capture changes in real-time, its focus isn't on providing immediate failover and an RPO of less than 1 second for high availability.

Overall Explanation:

A Multi-AZ deployment provides synchronous replication, ensuring minimal data loss (RPO less than 1 second) and automatic failover in case of an outage. Read replicas use asynchronous replication, which doesn't meet the RPO requirement. Autoscaling focuses on resource scaling, not data recovery. While DMS CDC captures changes, it doesn't provide the instant failover and data consistency offered by Multi-AZ. Therefore, Multi-AZ deployment is the only solution that guarantees the required RPO.

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Question 29

Not Attempted

Security



A company hosts a web application on Amazon EC2 instances within a private subnet of a VPC. An Application Load Balancer (ALB), spanning public subnets, distributes web traffic to these EC2 instances. For enhanced security, the company aims to restrict inbound traffic to the EC2 instances exclusively from the ALB, effectively preventing access from any other source, both internal and external to the private subnet. Which security configuration best achieves this?

- ☐ Configure a route in the route table to direct traffic from the internet to the private IP addresses of the EC2 instances.

This option is incorrect because it directly exposes the private EC2 instances to internet traffic, undermining the security requirements.

- ☒ Configure the security group for the EC2 instances to only allow traffic that comes from the security group for the ALB.

This is the correct solution. By configuring the EC2 instances' security group to accept traffic solely from the ALB's security group, you ensure that only traffic originating from the ALB can reach the instances, effectively blocking all other inbound traffic.

- ☐ Move the EC2 instances into the public subnet. Give the EC2 instances a set of Elastic IP addresses.

This option is incorrect because moving the instances to the public subnet and assigning Elastic IPs exposes them directly to the internet, contradicting the requirement to restrict access.

- ☐ Configure the security group for the ALB to allow any TCP traffic on any port.

This option is incorrect because configuring the ALB security group to allow all TCP traffic does not restrict access to the EC2 instances; it only controls what traffic can reach the ALB itself.

Overall Explanation:

The most secure approach is to configure the EC2 instances' security group to permit traffic only from the ALB's security group. This method ensures that only traffic routed through the ALB is allowed to reach the EC2 instances, effectively blocking all other attempts, whether internal or external, aligning with the company's security objective. Options A and C expose the instances directly to internet traffic, negating the security objective, and Option D only pertains to the ALB's security, not the EC2 instances.

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Question 30

Not Attempted

Storage



A research company currently runs experiments using a simulation application on Linux and a visualization application on Windows. The simulation application outputs intermediate data to an NFS share every 5 minutes. The visualization application displays the simulation output and requires an SMB file system. The company maintains two synchronized file systems, leading to data duplication and inefficient resource usage. The company wants to migrate these applications to AWS without modifying any code. Which solution will meet these requirements most effectively?

- ☐ **Migrate both applications to AWS Lambda. Create an Amazon S3 bucket to exchange data between the applications.**
This solution requires significant code changes to both applications to function within the Lambda environment and utilize S3 for data exchange, violating the 'no code changes' requirement.
- ☐ **Migrate both applications to Amazon Elastic Container Service (Amazon ECS). Configure Amazon FSx File Gateway for storage.**
This solution may require code changes depending on the existing application architecture and introduces FSx File Gateway, which primarily focuses on providing SMB access, not the required NFS access for the simulation application.
- ☐ **Migrate the simulation application to Linux Amazon EC2 instances. Migrate the visualization application to Windows EC2 instances. Configure Amazon Simple Queue Service (Amazon SQS) to exchange data between the applications.**
This solution requires significant architectural changes to use SQS for data transfer which goes against the requirements of not making any code changes to the applications.
- ☒ **Migrate the simulation application to Linux Amazon EC2 instances. Migrate the visualization application to Windows EC2 instances. Configure Amazon FSx for NetApp ONTAP for storage.**
This solution allows both applications to run on their respective operating systems within EC2, fulfilling the initial application requirements. FSx for NetApp ONTAP provides shared storage accessible via both NFS and SMB, eliminating data duplication and supporting the existing application functionalities without code changes.

Overall Explanation:

The most effective solution is to migrate the Linux simulation application to Linux EC2 instances and the Windows visualization application to Windows EC2 instances, utilizing Amazon FSx for NetApp ONTAP for storage. This setup provides a seamless migration path, supports both NFS and SMB protocols, eliminates data duplication, and requires no code changes. FSx for NetApp ONTAP ensures the data is synchronized and up to date across the file systems. This addresses the data duplication issues of the current setup.

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Question 31

Not Attempted

Management & Governance



As part of budget planning, management requires a report detailing AWS billed items, categorized by user. This data will be used to formulate department budgets. A solutions architect is tasked with identifying the most efficient method to obtain this report information. Which solution fulfills these requirements?

- ☐ Run a query with Amazon Athena to generate the report.
While Athena can process billing data, using Cost Explorer is a more direct and efficient approach for generating pre-formatted cost reports.
- ☒ Create a report in Cost Explorer and download the report.
Cost Explorer allows viewing, analyzing, and managing AWS costs and usage. It provides a variety of reports to track costs, including AWS billed items listed by user. Creating a report in Cost Explorer is a quick and easy way to obtain the information, and you can download the report in various formats.
- ☐ Access the bill details from the billing dashboard and download the bill.
Downloading the raw bill provides a comprehensive data set but requires significant processing and analysis to categorize expenses by user for budget planning purposes. Cost Explorer provides a pre-built reporting mechanism.
- ☐ Modify a cost budget in AWS Budgets to alert with Amazon Simple Email Service (Amazon SES).
AWS Budgets primarily focuses on setting budget thresholds and sending alerts when those thresholds are exceeded. It does not generate detailed reports of billed items listed by user for budget planning.

Overall Explanation:

Cost Explorer is the most efficient solution because it's designed to visualize and analyze AWS costs. It can readily generate reports breaking down costs by user, which directly addresses the requirement for department budget planning. Athena requires setting up and querying a database, and simply accessing the raw bill requires additional processing. Budgets is for monitoring spending against a defined amount, not generating detailed cost reports.

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Question 32

Not Attempted

[Compute](#)

A company hosts its static website by using Amazon S3. The company wants to add a contact form to its webpage. The contact form will have dynamic server-side components for users to input their name, email address, phone number, and user message. The company anticipates that there will be fewer than 100 site visits each month. Which solution will meet these requirements MOST cost-effectively?

- ☐ Host a dynamic contact form page in Amazon Elastic Container Service (Amazon ECS). Set up Amazon Simple Email Service (Amazon SES) to connect to any third-party email provider.

Using ECS for this small workload is overkill and not cost-effective.

- ☒ Create an Amazon API Gateway endpoint with an AWS Lambda backend that makes a call to Amazon Simple Email Service (Amazon SES).

API Gateway and Lambda are serverless and charge based on usage, making this the most cost-effective solution for a low-traffic contact form. SES is used for sending emails and has a generous free tier.

- ☐ Convert the static webpage to dynamic by deploying Amazon Lightsail. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.

Lightsail is more expensive than a serverless solution like API Gateway and Lambda for this workload, and WorkMail may also add unnecessary cost.

- ☐ Create a t2.micro Amazon EC2 instance. Deploy a LAMP (Linux, Apache, MySQL, PHP/Perl/Python) stack to host the webpage. Use client-side scripting to build the contact form. Integrate the form with Amazon WorkMail.

Running a t2.micro EC2 instance 24/7 is significantly more expensive than using API Gateway and Lambda for only 100 monthly visits, and WorkMail adds additional costs.

Overall Explanation:

The most cost-effective solution is to use Amazon API Gateway with an AWS Lambda backend that calls Amazon Simple Email Service (SES). This approach leverages serverless technologies, where you only pay for usage. With only 100 site visits per month, the costs associated with API Gateway and Lambda will be minimal, and Amazon SES has a free tier that likely covers the email sending costs. Options A, C, and D involve running infrastructure 24/7, which is less cost-effective for this small workload.

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Question 33

Not Attempted

Networking



A company hosts a static website on Amazon S3, served through Amazon CloudFront. Website updates pushed to the Git repository are deployed successfully to S3 via a CI/CD pipeline. However, website visitors do not see the latest changes. The pipeline's webhooks are correctly configured, and deployment messages confirm success. A solutions architect needs to ensure that website visitors see the updated content. What should the architect do?

☐ Add an Application Load Balancer.

An Application Load Balancer is used for distributing incoming application traffic across multiple targets, such as EC2 instances or containers. It does not address caching issues with static content.

☐ Add Amazon ElastiCache for Redis or Memcached to the database layer of the web application.

Adding ElastiCache improves database performance by caching frequently accessed data. However, the problem described is with static content not being updated, not the database performance.

☒ Invalidate the CloudFront cache.

Invalidating the CloudFront cache forces CloudFront to fetch the latest content from the origin (Amazon S3). This ensures that users see the most recent version of the website after updates have been deployed.

☐ Use AWS Certificate Manager (ACM) to validate the website's SSL certificate.

ACM is used to provision, manage, and deploy SSL/TLS certificates. While important for security, it does not address the issue of outdated content being served from the CloudFront cache.

Overall Explanation:

The core issue is that CloudFront is serving cached versions of the static website, even after updates have been deployed to S3. Invalidating the CloudFront cache will force it to retrieve the latest content from S3, resolving the problem. The other options do not address the caching problem. Therefore, invalidating the CloudFront cache is the correct solution.

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Question 34

Not Attempted

Database



A company wants to migrate a Windows-based application from on premises to the AWS Cloud. The application has three tiers: an application tier, a business tier, and a database tier with Microsoft SQL Server. The company wants to use specific features of SQL Server such as native backups and Data Quality Services. The company also needs to share files for processing between the tiers. How should a solutions architect design the architecture to meet these requirements?

- ☐ Host all three tiers on Amazon EC2 instances. Use Amazon FSx File Gateway for file sharing between the tiers.
FSx File Gateway is designed for low latency and efficient access to in-cloud FSx for Windows File Server file shares from your on-premises facility. It's not the best choice for file sharing within the AWS cloud.
- ☒ Host all three tiers on Amazon EC2 instances. Use Amazon FSx for Windows File Server for file sharing between the tiers.
This solution will allow the company to host all three tiers on Amazon EC2 instances while using Amazon FSx for Windows File Server to provide Windows-based file sharing between the tiers. This allows for native backups and Data Quality Services.
- ☐ Host the application tier and the business tier on Amazon EC2 instances. Host the database tier on Amazon RDS. Use Amazon Elastic File System (Amazon EFS) for file sharing between the tiers.

Amazon EFS supports the NFSv4.1 protocol and does not natively support the SMB protocol, and can't be used directly with Windows instances. Also, RDS does not allow native SQL server backups.

- ⊗ Host the application tier and the business tier on Amazon EC2 instances. Host the database tier on Amazon RDS. Use a Provisioned IOPS SSD (io2) Amazon Elastic Block Store (Amazon EBS) volume for file sharing between the tiers.

Amazon EBS is a block-level storage solution typically used for operating system level storage, not file sharing. Also, sharing an EBS volume between tiers doesn't make sense and RDS does not allow native SQL server backups.

Overall Explanation:

The best solution is to host all tiers on EC2 instances and use Amazon FSx for Windows File Server for file sharing. This allows the company to leverage specific SQL Server features like native backups and Data Quality Services, which are not available with RDS, and also provides SMB file sharing, essential for Windows-based applications. FSx File Gateway is intended for on-premises access to in-cloud FSx, and EFS does not support SMB natively.

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Question 35

Not Attempted

Storage



A company is migrating a group of Linux-based web servers to AWS. These web servers need to access files within a shared file store for certain content, and the company wants to avoid making any changes to the application code. What should a solutions architect implement to fulfill these requirements?

- ⊗ Create an Amazon S3 Standard bucket with access to the web servers.
Amazon S3 is an object storage service, not a file system. Mounting it as a file system typically requires application changes, which is not permitted in this scenario.
- ⊗ Configure an Amazon CloudFront distribution with an Amazon S3 bucket as the origin.
Amazon CloudFront is a content delivery network (CDN) and is not designed for providing shared file storage for web servers.
- ✓ Create an Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system on all web servers.
Amazon EFS is a fully managed NFS file system service that can be mounted on multiple EC2 instances simultaneously. It appears as a standard file system to the Linux web servers, requiring no

application changes.

- ⊗ Configure a General Purpose SSD (gp3) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume to all web servers.

Amazon EBS volumes can only be mounted to a single EC2 instance at a time, making them unsuitable for shared file storage across multiple web servers.

Overall Explanation:

Amazon EFS is the best solution because it provides a shared file system that can be mounted on multiple EC2 instances without any application code changes. S3 requires code changes, CloudFront is for content delivery, and EBS can only be mounted to one instance at a time. Therefore, EFS effectively addresses the requirement of providing a shared file store with no application modifications.

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Question 36

Not Attempted

Security



A company has an AWS Lambda function that needs read access to an Amazon S3 bucket located within the same AWS account. Which solution will meet these requirements while adhering to the principle of least privilege and providing the MOST secure access?

- ⊗ Apply an S3 bucket policy that grants read access to the S3 bucket.

While this grants access, it's less secure than using an IAM role for the Lambda function, as it directly manages access at the bucket level instead of assigning permissions to the function itself.

- ✔ Apply an IAM role to the Lambda function. Apply an IAM policy to the role to grant read access to the S3 bucket.

This is the most secure approach. It provides temporary credentials to the Lambda function through an IAM role, granting only the necessary permissions (read access) to the specific S3 bucket. This adheres to the principle of least privilege.

- ⊗ Embed an access key and a secret key in the Lambda function's code to grant the required IAM permissions for read access to the S3 bucket.

This is highly insecure. Embedding credentials in code is a security risk, as it makes them vulnerable to exposure.

- ⊗ Apply an IAM role to the Lambda function. Apply an IAM policy to the role to grant read access to all S3 buckets in the account.

While this uses an IAM role, it violates the principle of least privilege by granting access to all S3 buckets, instead of only the required one.

Overall Explanation:

The most secure and recommended approach is to assign an IAM role to the Lambda function and attach an IAM policy to the role that grants read access to the specific S3 bucket needed. This method avoids storing persistent credentials and limits the function's access to only the necessary resources, adhering to the principle of least privilege.

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Question 37

Not Attempted

Networking



A company provides a Voice over Internet Protocol (VoIP) service that uses UDP connections. The service consists of Amazon EC2 instances that run in an Auto Scaling group. The company has deployments across multiple AWS Regions. The company needs to route users to the Region with the lowest latency. The company also needs automated failover between Regions. Which solution will meet these requirements?

- ✔ Deploy a Network Load Balancer (NLB) and an associated target group. Associate the target group with the Auto Scaling group. Use the NLB as an AWS Global Accelerator endpoint in each Region.

This solution leverages NLB for UDP support, Auto Scaling groups for scalability, and Global Accelerator for low latency routing and automatic regional failover.

- ⊗ Deploy an Application Load Balancer (ALB) and an associated target group. Associate the target group with the Auto Scaling group. Use the ALB as an AWS Global Accelerator endpoint in each Region.

ALB does not support UDP, therefore it is not suitable for VoIP traffic.

- ⊗ Deploy a Network Load Balancer (NLB) and an associated target group. Associate the target group with the Auto Scaling group. Create an Amazon Route 53 latency record that points to aliases for each NLB. Create an Amazon CloudFront distribution that uses the latency record as an origin.

CloudFront is designed for caching content and is not suitable for real-time communication using UDP. While Route 53 latency routing helps, CloudFront isn't the right fit for VoIP.

- ⊗ Deploy an Application Load Balancer (ALB) and an associated target group. Associate the target group with the Auto Scaling group. Create an Amazon Route 53 weighted record that points to aliases for each ALB. Deploy an Amazon CloudFront distribution that uses the weighted record as an origin.

ALB does not support UDP, and CloudFront is not designed for real-time UDP traffic. Weighted records in Route 53 do not inherently provide latency-based routing.

Overall Explanation:

The correct solution is to use a Network Load Balancer (NLB) with an associated target group linked to the Auto Scaling group, and then use the NLB as an AWS Global Accelerator endpoint in each Region. NLB supports UDP connections required for VoIP. Global Accelerator provides low latency routing and automatic regional failover. ALB does not support UDP. CloudFront is not intended for real-time traffic.

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Question 38

Not Attempted

Compute



A company hosts a web application on multiple Amazon EC2 instances. The EC2 instances are in an Auto Scaling group that scales in response to user demand. The company wants to optimize cost savings without making a long-term commitment. Which EC2 instance purchasing option should a solutions architect recommend to meet these requirements?

- ⊗ Dedicated Instances only

Dedicated Instances are the most expensive option and don't provide cost savings without long-term commitment.

- ⊗ On-Demand Instances only

On-Demand Instances are flexible but more expensive than Spot Instances, making them less optimal for cost savings without a commitment.

- ✓ A mix of On-Demand Instances and Spot Instances

A mix of On-Demand and Spot Instances allows for cost optimization. On-Demand provides a baseline, and Spot provides cost savings for variable demand, without a long-term commitment.

⊗ A mix of On-Demand Instances and Reserved Instances

Reserved Instances require a commitment of 1 or 3 years, which is not suitable when the company wants to avoid long-term commitments.

Overall Explanation:

Using a mix of On-Demand and Spot Instances in an Auto Scaling group allows the company to leverage the availability of On-Demand Instances for baseline capacity and the cost savings of Spot Instances for peak demand or non-critical workloads. This approach provides optimal cost savings without requiring a long-term commitment, as Spot Instances can be terminated if prices increase beyond a set bid price. The Auto Scaling group can be configured to automatically replace terminated Spot Instances with other Spot Instances or On-Demand Instances.

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Question 39

Not Attempted

Security



A media company uses Amazon CloudFront for its publicly available streaming video content hosted in Amazon S3. The company needs to secure the video content by controlling access. Some users employ a custom HTTP client that doesn't support cookies, while others can't modify the hardcoded URLs they use for access. Which two services or methods will meet these requirements with the LEAST impact on users?

✓ Signed cookies

Signed cookies can control access for users who cannot change their hardcoded URLs. CloudFront uses the cookie to authenticate access to the content.

✓ Signed URLs

Signed URLs provide temporary access to S3 objects and work for users with custom HTTP clients that don't support cookies, as the authentication is embedded in the URL.

⊗ AWS AppSync

AWS AppSync is a GraphQL service and not directly relevant to securing static video content in S3.

⊗ JSON Web Token (JWT)

JWTs can be used to authorize access, but typically require integration with Lambda@Edge or similar to validate and control access to CloudFront content, adding complexity.

⊗ AWS Secrets Manager

AWS Secrets Manager is used for storing secrets and is not directly related to securing video content delivery in this scenario.

⊗ N/A

N/A

Overall Explanation:

Signed Cookies and Signed URLs are the best choices. Signed URLs cater to users with HTTP clients lacking cookie support by embedding authentication in the URL. Signed Cookies address the issue of unchangeable hardcoded URLs by allowing authentication via cookies. These methods minimize user impact and meet the requirements effectively. JWT would work but is not the simplest solution.

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Question 40

Not Attempted

Database



A company is designing a real-time data analytics platform. The platform will ingest streaming data from various sources, transform the data, and store it in Amazon S3. Data analysts must be able to use SQL to query the transformed data in S3 for reporting purposes. Which combination of services will meet these requirements with minimal operational overhead? (Choose two.)

- ✓ Use Amazon Kinesis Data Streams to ingest the data. Use Amazon Kinesis Data Analytics to transform the data. Use Amazon Kinesis Data Firehose to load the data into Amazon S3. Use Amazon Athena to query the data in Amazon S3.

This solution leverages Kinesis Data Streams, Data Analytics, and Firehose for streaming, transformation, and loading, respectively. Athena provides the SQL querying capability on S3.

- ✓ Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to ingest the data. Use AWS Glue to transform the data and load the data into Amazon S3. Use Amazon Athena to query the data in Amazon S3.

This solution uses MSK for streaming, AWS Glue for transformation and S3 loading, and Athena for querying. AWS Glue supports streaming ETL jobs from sources like MSK.

- ⊗ Use AWS Database Migration Service (AWS DMS) to ingest the data. Use Amazon EMR to transform the data and load the data into Amazon S3. Use Amazon Athena to query the data in Amazon S3.

DMS is not optimized for real-time streaming data ingestion. EMR introduces more operational overhead than Kinesis Data Analytics or Glue for this use case.

- ⊗ Use Amazon Managed Streaming for Apache Kafka (Amazon MSK) to ingest the data. Use Amazon Kinesis Data Analytics to transform the data and load the data into Amazon S3. Use the Amazon RDS query editor to query the data in Amazon S3.

The RDS query editor is not designed for querying data directly in S3. Athena is the appropriate service for querying S3 data using SQL.

- ⊗ Use Amazon Kinesis Data Streams to ingest the data. Use AWS Glue to transform the data. Use Amazon Kinesis Data Firehose to load the data into Amazon S3. Use the Amazon RDS query editor to query the transformed data from Amazon S3.

The RDS query editor is not designed for querying data directly in S3. Athena is the appropriate service for querying S3 data using SQL.

Overall Explanation:

The most suitable solutions are using Kinesis Data Streams, Data Analytics, Firehose, and Athena (Option A), and using MSK, AWS Glue, and Athena (Option B). Both meet the requirements of real-time ingestion, transformation, S3 storage, and SQL querying. Option A leverages Kinesis services, while Option B uses MSK and Glue. The other options are incorrect because DMS is not ideal for streaming data, EMR adds unnecessary operational overhead, and the RDS query editor cannot query S3 data.

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Question 41

Not Attempted

Storage



A company's on-premises volume backup solution is reaching its end of life. The company wants to adopt AWS as part of a new backup strategy, needing to maintain local access to all data while it's backed up on AWS. They require the data to be automatically and securely transferred to AWS. Which solution satisfies these requirements?

- ⊗ Use AWS Snowball to migrate data out of the on-premises solution to Amazon S3. Configure on-premises systems to mount the Snowball S3 endpoint to provide local access to the data.
Snowball is primarily for initial migration and not for continuous, automated backup and local access.
- ⊗ Use AWS Snowball Edge to migrate data out of the on-premises solution to Amazon S3. Use the Snowball Edge file interface to provide on-premises systems with local access to the data.
Snowball Edge is useful for edge computing and data migration, but not designed for continuous, automated backup with maintained local access.
- ⊗ Use AWS Storage Gateway and configure a cached volume gateway. Run the Storage Gateway software appliance on premises and configure a percentage of data to cache locally. Mount the gateway storage volumes to provide local access to the data.
A cached volume gateway only stores frequently accessed data locally, not all data as the requirement states. The primary data is stored in S3.
- ✔ Use AWS Storage Gateway and configure a stored volume gateway. Run the Storage Gateway software appliance on premises and map the gateway storage volumes to on-premises storage. Mount the gateway storage volumes to provide local access to the data.
A stored volume gateway stores the primary data locally and asynchronously backs it up to AWS, meeting the requirements of local access to all data and automated, secure transfer to AWS.

Overall Explanation:

The stored volume gateway configuration of AWS Storage Gateway fulfills all the requirements. It provides local access to all the data on-premises while asynchronously backing up the data to AWS securely and automatically. Cached volumes do not keep all data locally. Snowball and Snowball Edge are primarily for data migration, not continuous backup.

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Question 42

Not Attempted

Networking



An application is hosted on Amazon EC2 instances within a VPC. These instances require access to an Amazon S3 bucket. To ensure traffic does not traverse the public internet, how should a solutions architect configure the access?

- ☐ Create a private hosted zone by using Amazon Route 53.
A private hosted zone in Route 53 is used for internal DNS resolution within a VPC but does not provide a private network path to S3.
- ☒ Set up a gateway VPC endpoint for Amazon S3 in the VPC.
A gateway VPC endpoint for S3 provides private connectivity from instances within a VPC to S3 without requiring internet access.
- ☐ Configure the EC2 instances to use a NAT gateway to access the S3 bucket.
A NAT gateway allows instances in a private subnet to connect to the internet. This option would not prevent traffic from traversing the internet.
- ☐ Establish an AWS Site-to-Site VPN connection between the VPC and the S3 bucket.
AWS Site-to-Site VPN is used to connect your on-premises network to your VPC, not to connect your VPC to S3.

Overall Explanation:

A gateway VPC endpoint for Amazon S3 is the correct solution. It enables EC2 instances within a VPC to access S3 privately, without routing traffic over the internet. S3 and DynamoDB are the only services with Gateway endpoint options.

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Question 43

Not Attempted

Security



An e-commerce company stores terabytes of customer data, including PII, in the AWS Cloud. This data will be used by three different applications. However, only one application requires access to the PII. The PII needs to be removed before the other two applications process the data. Which solution meets these requirements with the LEAST operational overhead?

- ☐ Store the data in an Amazon DynamoDB table. Create a proxy application layer to intercept and process the data that each application requests.

Creating a proxy application layer adds complexity and operational overhead for managing and maintaining the proxy.

- ☒ Store the data in an Amazon S3 bucket. Process and transform the data by using S3 Object Lambda before returning the data to the requesting application.
S3 Object Lambda allows for on-the-fly data transformation and PII redaction as data is retrieved, minimizing operational overhead by avoiding multiple copies or custom applications.
- ☐ Process the data and store the transformed data in three separate Amazon S3 buckets so that each application has its own custom dataset. Point each application to its respective S3 bucket.
Storing three copies of the data in separate buckets increases storage costs and operational overhead for managing and updating the data.
- ☐ Process the data and store the transformed data in three separate Amazon DynamoDB tables so that each application has its own custom dataset. Point each application to its respective DynamoDB table.
Using DynamoDB for all data and maintaining multiple copies significantly increases costs and operational overhead, especially for terabytes of data.

Overall Explanation:

Storing the data in S3 and using S3 Object Lambda offers the least operational overhead. It allows you to process and transform the data only when it's being retrieved, removing PII for applications that don't need it without creating multiple copies of the data. This solution centralizes data storage and transformation logic, reducing management and storage costs. S3 Object Lambda enables you to apply custom code to your data retrieval requests, allowing you to remove PII before returning the data to the requesting application.

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Question 44

Not Attempted

Networking



A development team has launched a new application that is hosted on Amazon EC2 instances inside a development VPC. A solutions architect needs to create a new VPC in the same account. The new VPC will be peered with the development VPC. The VPC CIDR block for the development VPC is 192.168.0.0/24. The solutions architect needs to create a CIDR block for the

new VPC. The CIDR block must be valid for a VPC peering connection to the development VPC. What is the SMALLEST CIDR block that meets these requirements?

☐ 10.0.1.0/32

A /32 CIDR block represents a single host, not a valid range for a VPC.

☐ 192.168.0.0/24

This CIDR block overlaps with the existing development VPC's CIDR block, making it invalid for peering.

☐ 192.168.1.0/32

A /32 CIDR block represents a single host, not a valid range for a VPC.

☒ 10.0.1.0/24

This CIDR block is a valid, non-overlapping CIDR block that can be used for the new VPC.

Overall Explanation:

The CIDR block for the new VPC must not overlap with the existing 192.168.0.0/24 block of the development VPC. The options 10.0.1.0/32 and 192.168.1.0/32 are invalid as they are /32, meaning single IP addresses rather than a CIDR block. 192.168.0.0/24 is invalid because it overlaps with the original development VPC. Thus, 10.0.1.0/24 is the smallest valid CIDR block.

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Question 45

Not Attempted

Compute



A company has deployed an application on five Amazon EC2 instances behind an Application Load Balancer (ALB). The ALB distributes traffic to the instances through a target group. The average CPU utilization on each instance is consistently below 10%, but experiences occasional spikes up to 65%. A solutions architect needs to implement a solution to automatically scale the application. The solution must minimize costs while ensuring sufficient CPU resources during traffic surges. Which solution meets these requirements?

☐ Create an Amazon CloudWatch alarm that enters the ALARM state when the CPUUtilization metric is less than 20%. Create an AWS Lambda function that the

CloudWatch alarm invokes to terminate one of the EC2 instances in the ALB target group.

This approach only addresses low CPU utilization and does not ensure sufficient resources during surges. Terminating instances when CPU is low doesn't solve high CPU load issues.

- ✔ Create an EC2 Auto Scaling group. Select the existing ALB as the load balancer and the existing target group as the target group. Set a target tracking scaling policy that is based on the ASGAverageCPUUtilization metric. Set the minimum instances to 2, the desired capacity to 3, the maximum instances to 6, and the target value to 50%. Add the EC2 instances to the Auto Scaling group.

This solution leverages an Auto Scaling group to dynamically adjust the number of EC2 instances based on CPU utilization. Target tracking ensures that the average CPU stays around 50%, scaling up during surges and down during low utilization, optimizing cost. Integrating with the existing ALB and target group minimizes changes.

- ✘ Create an EC2 Auto Scaling group. Select the existing ALB as the load balancer and the existing target group as the target group. Set the minimum instances to 2, the desired capacity to 3, and the maximum instances to 6. Add the EC2 instances to the Auto Scaling group.

This solution creates an Auto Scaling group but lacks a defined scaling policy based on CPU utilization. Without a scaling policy, the group will not automatically adjust the number of instances in response to CPU spikes, failing to address the requirements effectively.

- ✘ Create two Amazon CloudWatch alarms. Configure the first CloudWatch alarm to enter the ALARM state when the average CPUUtilization metric is below 20%. Configure the second CloudWatch alarm to enter the ALARM state when the average CPUUtilization metric is above 50%. Configure the alarms to publish to an Amazon Simple Notification Service (Amazon SNS) topic to send an email message. After receiving the message, log in to decrease or increase the number of EC2 instances that are running.

This approach relies on manual intervention, which is not cost-effective and doesn't meet the requirement for automated scalability. It involves manually adjusting the number of instances based on email notifications, which is not ideal for handling rapid surges.

Overall Explanation:

The correct answer is B. This solution utilizes an EC2 Auto Scaling group configured with a target tracking scaling policy based on ASGAverageCPUUtilization. This enables the application to automatically scale the number of EC2 instances between a minimum of 2 and a maximum of 6 to maintain an average CPU utilization of 50%. This cost-effectively ensures the application has enough CPU resources during surges without over-

provisioning when resources are not needed. The existing ALB and target group are integrated into the Autoscaling group without any architectural changes.

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Question 46

Not Attempted

Networking



A company is running a critical business application on Amazon EC2 instances behind an Application Load Balancer. The EC2 instances run in an Auto Scaling group and access an Amazon RDS DB instance. The current architecture places all resources in a single Availability Zone, failing an operational review. To improve high availability, a solutions architect must update the design to use a second Availability Zone. Which solution will achieve this?

- ☐ Provision a subnet in each Availability Zone. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance with connections to each network.

This option is incorrect because directly connecting the DB instance to each network is not the recommended approach for high availability.

- ☐ Provision two subnets that extend across both Availability Zones. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance with connections to each network.

This option is incorrect because a subnet cannot span multiple Availability Zones.

- ☒ Provision a subnet in each Availability Zone. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance for Multi-AZ deployment.

This option is correct. It places EC2 instances in multiple AZs for compute availability and configures the RDS DB instance for Multi-AZ deployment for database availability.

- ☐ Provision a subnet that extends across both Availability Zones. Configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones. Configure the DB instance for Multi-AZ deployment.

This option is incorrect because a subnet cannot extend across multiple Availability Zones.

Overall Explanation:

The correct answer is to provision a subnet in each Availability Zone, configure the Auto Scaling group to distribute the EC2 instances across both Availability Zones, and configure the DB instance for Multi-AZ deployment. Subnets are confined to single AZs, and Auto Scaling groups can span multiple AZs. Multi-AZ deployment for the RDS instance provides failover capabilities to another AZ. Configuring the DB instance for Multi-AZ deployment ensures that a standby replica of the DB instance is available in another Availability Zone, providing redundancy and failover capabilities.

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Question 47

Not Attempted

Storage



A research laboratory needs to process approximately 8 TB of data. The laboratory requires sub-millisecond latencies and a minimum throughput of 6 GBps for the storage subsystem. Hundreds of Amazon EC2 instances that run Amazon Linux will distribute and process the data. Which solution will meet the performance requirements?

- ☐ Create an Amazon FSx for NetApp ONTAP file system. Set each volume's tiering policy to ALL. Import the raw data into the file system. Mount the file system on the EC2 instances.

FSx for NetApp ONTAP may not meet the throughput and latency requirements, especially with tiering set to ALL.

- ☒ Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent SSD storage. Select the option to import data from and export data to Amazon S3. Mount the file system on the EC2 instances.

FSx for Lustre with SSD provides sub-millisecond latencies and high throughput, meeting the requirements. S3 provides cost effective storage and the import/export feature simplifies data management.

- ☐ Create an Amazon S3 bucket to store the raw data. Create an Amazon FSx for Lustre file system that uses persistent HDD storage. Select the option to import data from and export data to Amazon S3. Mount the file system on the EC2 instances.

FSx for Lustre with HDD will not meet the sub-millisecond latency requirement.

- ☐ Create an Amazon FSx for NetApp ONTAP file system. Set each volume's tiering policy to NONE. Import the raw data into the file system. Mount the file system on the EC2

instances.

FSx for NetApp ONTAP, even with tiering set to NONE, is unlikely to meet the 6 GBps throughput requirement.

Overall Explanation:

Amazon FSx for Lustre with SSD storage provides the necessary sub-millisecond latencies and a minimum throughput of 6 GBps. Using persistent SSD ensures data durability, and importing/exporting data to/from Amazon S3 allows for cost-effective storage and easy data management.

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Question 48

Not Attempted

Security



A company uses AWS Organizations to manage multiple AWS accounts for different departments. The management account has an Amazon S3 bucket that contains project reports. The company wants to limit access to this S3 bucket to only users of accounts within the organization in AWS Organizations. Which solution meets these requirements with the LEAST amount of operational overhead?

- ☒ Add the `aws:PrincipalOrgID` global condition key with a reference to the organization ID to the S3 bucket policy.

This option allows you to specify the organization ID in the S3 bucket policy, restricting access to only users within that organization. It avoids managing individual accounts or OUs, resulting in the least operational overhead.

- ☐ Create an organizational unit (OU) for each department. Add the `aws:PrincipalOrgPaths` global condition key to the S3 bucket policy.

While this approach can limit access, it requires creating and managing OUs, increasing operational overhead compared to using `aws:PrincipalOrgID`.

- ☐ Use AWS CloudTrail to monitor the `CreateAccount`, `InviteAccountToOrganization`, `LeaveOrganization`, and `RemoveAccountFromOrganization` events. Update the S3 bucket policy accordingly.

CloudTrail only logs API events; it doesn't directly control access. Implementing this would require complex automation to update the S3 bucket policy based on CloudTrail logs, leading to high operational overhead and potential delays.

- ⊗ Tag each user that needs access to the S3 bucket. Add the `aws:PrincipalTag` global condition key to the S3 bucket policy.

Tagging individual users is not scalable or efficient for managing access based on organizational membership. It introduces significant operational overhead as user tags need to be maintained.

Overall Explanation:

The correct solution is to add the `aws:PrincipalOrgID` global condition key to the S3 bucket policy. This allows you to control access to the S3 bucket based on the organization ID. This approach is the most efficient and requires the least amount of operational overhead because AWS Organizations already tracks the organization ID of each account, so you can simply add the `aws:PrincipalOrgID` condition key to the S3 bucket policy and reference the organization ID. Alternatives involve managing OUs, tagging users, or using CloudTrail to indirectly control access, all of which increase operational complexity.

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Question 49

Not Attempted

Database



A development team runs monthly resource-intensive tests on its general purpose Amazon RDS for MySQL DB instance with Performance Insights enabled. The testing lasts for 48 hours once a month and is the only process that uses the database. The team wants to reduce the cost of running the tests without reducing the compute and memory attributes of the DB instance. Which solution meets these requirements MOST cost-effectively?

- ⊗ Stop the DB instance when tests are completed. Restart the DB instance when required.
Stopping the instance still incurs charges for storage and backups. Additionally, the instance may automatically restart after seven days, which is not ideal for a monthly testing schedule.
- ⊗ Use an Auto Scaling policy with the DB instance to automatically scale when tests are completed.
Auto Scaling is not applicable to RDS instances in the same way as EC2 instances and will not effectively reduce costs in this scenario.
- ✓ Create a snapshot when tests are completed. Terminate the DB instance and restore the snapshot when required.

This option offers the most significant cost savings by eliminating instance charges when not in use. Snapshot storage is cheaper than maintaining a stopped instance. When needed, the instance can be quickly restored from the snapshot.

- ⊗ Modify the DB instance to a low-capacity instance when tests are completed. Modify the DB instance again when required.

Modifying the instance size is not as cost-effective as terminating the instance and using snapshots, and involves downtime for modification.

Overall Explanation:

Creating a snapshot and terminating the instance is the MOST cost-effective solution.

When the monthly tests are completed, a snapshot is created. Then, the RDS instance is terminated, thus eliminating compute costs. Storage costs for the snapshot are significantly less than maintaining a running or stopped instance. When required again, the DB instance can be restored from the snapshot. Stopping the instance still incurs storage costs, so C is the better option.

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Question 50

Not Attempted

Database



A company needs to migrate a legacy application from an on-premises data center to the AWS Cloud due to hardware capacity constraints. The application runs 24/7, and the application's database storage is continuously growing. To meet these requirements most cost-effectively, what should a Solutions Architect recommend?

- ⊗ Migrate the application layer to Amazon EC2 Spot Instances. Migrate the data storage layer to Amazon S3.

EC2 Spot Instances are not suitable for 24/7 applications because they can be terminated with short notice. Amazon S3 is object storage and not suitable for a relational database.

- ⊗ Migrate the application layer to Amazon EC2 Reserved Instances. Migrate the data storage layer to Amazon RDS On-Demand Instances.

While Reserved Instances are a good choice for the application layer, On-Demand RDS instances can be more expensive in the long run for a database that is constantly growing. Reserved Aurora is a better choice

- ☒ Migrate the application layer to Amazon EC2 Reserved Instances. Migrate the data storage layer to Amazon Aurora Reserved Instances.

This is the most cost-effective solution. Reserved Instances for EC2 provide cost savings for steady-state workloads. Aurora Reserved Instances offer cost benefits and scalability for growing database storage.

- ☐ Migrate the application layer to Amazon EC2 On-Demand Instances. Migrate the data storage layer to Amazon RDS Reserved Instances.

On-Demand EC2 instances are expensive for 24/7 applications. RDS Reserved Instances are a good option, but Aurora reserved will be cheaper overall

Overall Explanation:

The most cost-effective approach involves using Reserved Instances for both the application (EC2) and database (Aurora) layers. Reserved Instances offer significant cost savings compared to On-Demand instances for continuous operation. Aurora provides scalability for the growing database storage. Using spot instances are not a good option for 24/7 applications.

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Question 51

Not Attempted

Storage



A university research laboratory needs to migrate 30 TB of data from an on-premises Windows file server to Amazon FSx for Windows File Server. The laboratory has a 1 Gbps network link that many other departments in the university share. The laboratory wants to implement a data migration service that will maximize the performance of the data transfer. However, the laboratory needs to be able to control the amount of bandwidth that the service uses to minimize the impact on other departments. The data migration must take place within the next 5 days. Which AWS solution will meet these requirements?

- ☐ AWS Snowcone

AWS Snowcone is a physical device and delivering the device will take 4-6 days, making it unsuitable for migration within 5 days. Also, it is not designed for controlled bandwidth usage over a network.

- ☐ Amazon FSx File Gateway

Amazon FSx File Gateway primarily focuses on providing low-latency access to FSx file systems from on-premises environments and is not suitable for large-scale data migration.

✔ AWS DataSync

AWS DataSync is designed for online data transfer and allows you to control the amount of bandwidth used during data transfer, schedule data transfers, and configure transfer rates. It can copy large amounts of data between on-premises storage and Amazon FSx for Windows File Server at high speeds.

✘ AWS Transfer Family

AWS Transfer Family is a fully managed service that enables the transfer of files over SFTP, FTPS, and FTP directly into and out of Amazon S3 or Amazon EFS. It is not the ideal service for migrating data to Amazon FSx for Windows File Server, especially with the need for bandwidth control.

Overall Explanation:

AWS DataSync is the correct choice because it's designed for efficient online data transfer with built-in bandwidth throttling. This addresses the need to migrate 30TB of data within 5 days while minimizing the impact on other departments sharing the 1 Gbps network link. Snowcone is unsuitable due to shipping time, FSx File Gateway is for access rather than migration, and Transfer Family isn't optimal for FSx for Windows File Server migrations.

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Question 52

Not Attempted

Storage



A company is developing a mobile application for streaming slow-motion videos. Currently, the application captures video clips and uploads them in raw format to an Amazon S3 bucket. Users are experiencing buffering and playback issues due to the large file sizes. The company wants to improve performance and scalability while minimizing operational overhead. Which combination of solutions will best address these requirements? (Choose two.)

✔ Deploy Amazon CloudFront for content delivery and caching.

Amazon CloudFront, a CDN, improves app performance and scalability by caching content at edge locations, reducing latency, and optimizing video delivery. It also provides DDoS protection, SSL/TLS encryption, and content compression.

✘ Use AWS DataSync to replicate the video files across AWS Regions in other S3 buckets.

AWS DataSync helps to replicate data between on-premises storage and AWS, between AWS storage services, as well as across AWS Regions. This solution is not the optimal choice for streaming the video in a cost-effective way.

✔ Use Amazon Elastic Transcoder to convert the video files to more appropriate formats.

Amazon Elastic Transcoder optimizes video formats for mobile devices, reducing file sizes and improving playback performance. It can also convert videos into multiple formats to support different devices.

✘ Deploy an Auto Scaling group of Amazon EC2 instances in Local Zones for content delivery and caching.

While EC2 instances can be configured for content delivery and caching, this approach involves more operational overhead compared to using a managed service like CloudFront.

✘ Deploy an Auto Scaling group of Amazon EC2 instances to convert the video files to more appropriate formats.

While this could work, Elastic Transcoder is a managed service and will be much less overhead.

Overall Explanation:

The most effective solution involves using Amazon CloudFront for caching and content delivery (Option 1) in combination with Amazon Elastic Transcoder to optimize video formats (Option 3). CloudFront reduces latency and improves delivery, while Elastic Transcoder reduces file sizes and ensures compatibility across devices, both of which minimize operational overhead.

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Question 53

Not Attempted

Compute



A company is launching a new application deployed on an Amazon Elastic Container Service (Amazon ECS) cluster using the Fargate launch type. The company is monitoring CPU and memory usage due to anticipated high traffic upon launch. To optimize costs as utilization decreases, which solution should a solutions architect recommend?

- ⊗ Use Amazon EC2 Auto Scaling to scale at certain periods based on previous traffic patterns.

This is incorrect because the application is running on Fargate, not EC2 instances. EC2 Auto Scaling cannot directly scale Fargate tasks.

- ⊗ Use an AWS Lambda function to scale Amazon ECS based on metric breaches that trigger an Amazon CloudWatch alarm.

This is incorrect because, while possible, using a Lambda function for scaling is overly complex. AWS Application Auto Scaling provides a more straightforward and managed approach.

- ⊗ Use Amazon EC2 Auto Scaling with simple scaling policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm.

This is incorrect because the application is running on Fargate, not EC2 instances. EC2 Auto Scaling cannot directly scale Fargate tasks.

- ✔ Use AWS Application Auto Scaling with target tracking policies to scale when ECS metric breaches trigger an Amazon CloudWatch alarm.

This is correct. AWS Application Auto Scaling with target tracking policies allows ECS to automatically scale based on metrics like CPU and memory utilization. Target tracking dynamically adjusts the number of tasks to maintain the desired utilization level, optimizing costs and performance.

Overall Explanation:

AWS Application Auto Scaling with target tracking policies is the most suitable solution. It allows the ECS cluster to automatically scale in or out based on predefined metrics (e.g., CPU utilization) monitored by CloudWatch. This approach ensures that the application can handle high traffic loads while also reducing costs during periods of low utilization. EC2 Auto Scaling is not applicable for Fargate, and using Lambda for scaling would be an unnecessarily complex solution. Target tracking policies automatically adjust the resources to meet the target utilization.

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Question 54

Not Attempted

Storage



A company recently established a disaster recovery site in a separate AWS Region. They need a solution to frequently transfer large datasets between NFS file systems located in both Regions.

The priority is to minimize ongoing management and administrative tasks.

✔ Use AWS DataSync.

AWS DataSync is a fully managed service designed for efficient and automated data transfers between on-premises storage and AWS, as well as between AWS services in different Regions. It simplifies the process and reduces operational overhead.

✘ Use AWS Snowball devices.

AWS Snowball is primarily designed for large, one-time data migrations or for transferring data from edge locations to AWS, not for frequent, automated transfers between Regions.

✘ Set up an SFTP server on Amazon EC2.

Setting up an SFTP server on EC2 requires manual configuration, management, and scaling, leading to increased operational overhead.

✘ Use AWS Database Migration Service (AWS DMS).

AWS DMS is designed for database migrations, not for transferring data between NFS file systems.

Overall Explanation:

AWS DataSync is the most suitable solution because it's specifically designed for efficient, automated, and managed data transfers between NFS file systems across AWS Regions. It minimizes operational overhead compared to other solutions by automating the data transfer process, providing monitoring capabilities, and ensuring data integrity.

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Question 55

Not Attempted

Storage



A company is designing a shared storage solution for a gaming application hosted in the AWS Cloud. The company requires SMB protocol support for client access to the data, and the solution must be fully managed, minimizing administrative overhead. Which AWS service best meets these requirements?

✘ Create an AWS DataSync task that shares the data as a mountable file system. Mount the file system to the application server.

AWS DataSync is primarily a data transfer service and does not provide a fully managed SMB file system. It's used for moving data between on-premises and AWS storage services. It does not

provide the SMB access required in a fully managed fashion.

- ⊗ Create an Amazon EC2 Windows instance. Install and configure a Windows file share role on the instance. Connect the application server to the file share.

While this approach provides SMB access, it is not a fully managed solution. The company would be responsible for managing the EC2 instance, OS patching, file share configuration, and backups, increasing operational overhead.

- ✓ Create an Amazon FSx for Windows File Server file system. Attach the file system to the origin server. Connect the application server to the file system.

Amazon FSx for Windows File Server is a fully managed native Windows file system that provides SMB access. It handles the underlying infrastructure, including provisioning, setup, maintenance, backups, and security, making it the ideal solution for a fully managed SMB file share.

- ⊗ Create an Amazon S3 bucket. Assign an IAM role to the application to grant access to the S3 bucket. Mount the S3 bucket to the application server.

Amazon S3 is an object storage service, not a file system. While it offers high scalability and availability, it doesn't natively support the SMB protocol. Mounting S3 as a file system typically involves using third-party tools, which doesn't provide native SMB access and adds complexity.

Overall Explanation:

Amazon FSx for Windows File Server provides a fully managed, native Windows file system accessible via the SMB protocol, which fulfills the requirements of providing SMB access to data and a fully managed solution. FSx handles administrative tasks such as provisioning, setup, maintenance, file share management, backups, security, and software patching. This reduces operational overhead, making it suitable for the company's needs. AWS DataSync is used for data transfer, EC2 with Windows File Share requires manual management, and S3 is object storage that doesn't natively support SMB.

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Question 56

Not Attempted

Networking



A company wants to run an in-memory database for a latency-sensitive application that processes more than 100,000 transactions each minute on Amazon EC2 instances. The application requires high network throughput. A solutions architect needs to provide a cost-

effective network design that minimizes data transfer charges. Which solution meets these requirements?

- ☒ Launch all EC2 instances in the same Availability Zone within the same AWS Region. Specify a placement group with cluster strategy when launching EC2 instances.
Launching instances within a single AZ and using a cluster placement group provides the lowest network latency and highest bandwidth between instances. This maximizes performance and minimizes data transfer charges as communication within the same AZ and placement group is free.
- ☐ Launch all EC2 instances in different Availability Zones within the same AWS Region. Specify a placement group with partition strategy when launching EC2 instances.
Launching instances across multiple Availability Zones increases network latency and introduces data transfer charges between AZs. Partition placement groups do not provide the low latency required.
- ☐ Deploy an Auto Scaling group to launch EC2 instances in different Availability Zones based on a network utilization target.
Auto Scaling across different Availability Zones increases network latency and introduces data transfer charges between AZs. This does not guarantee the lowest latency or minimize cost.
- ☐ Deploy an Auto Scaling group with a step scaling policy to launch EC2 instances in different Availability Zones.
Auto Scaling across different Availability Zones increases network latency and introduces data transfer charges between AZs. This does not meet the latency or cost requirements.

Overall Explanation:

The optimal solution is to launch all EC2 instances in the same Availability Zone and use a cluster placement group. This minimizes network latency and data transfer charges, and provides the required high network throughput. Cluster placement groups pack instances close together inside an Availability Zone, enabling workloads to achieve low-latency network performance.

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Question 57

Not Attempted

Storage



A company currently hosts its application servers on-premises and is planning a migration to AWS. To minimize the need for scaling its on-premises Internet Small Computer Systems Interface (iSCSI) storage, the company wants to ensure that only recently accessed data remains stored locally. The remaining data should be stored in AWS. Which AWS solution should the company implement to meet these requirements?

☐ Amazon S3 File Gateway

Amazon S3 File Gateway provides file-based access to S3. It doesn't address the need to keep recently accessed iSCSI data locally.

☐ AWS Storage Gateway Tape Gateway

AWS Storage Gateway Tape Gateway is designed for virtual tape backup and archival in AWS, not for general-purpose iSCSI storage migration and local caching.

☐ AWS Storage Gateway Volume Gateway stored volumes

AWS Storage Gateway Volume Gateway stored volumes store the entire dataset on-premises and asynchronously back up to AWS. This does not meet the requirement of keeping only recently accessed data locally.

☒ AWS Storage Gateway Volume Gateway cached volumes

AWS Storage Gateway Volume Gateway cached volumes store recently accessed data on-premises as a cache, while the entire dataset is stored in Amazon S3. This meets the requirement of minimizing on-premises storage while keeping frequently accessed data local for low-latency access.

Overall Explanation:

AWS Storage Gateway Volume Gateway cached volumes are the best solution. They allow the company to keep frequently accessed data on-premises for low latency, while less frequently accessed data is stored in Amazon S3. This minimizes the need to scale on-premises iSCSI storage while still providing access to all data. The other options do not fulfill the requirement of keeping only recently accessed data locally and storing the rest in AWS.

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A company has multiple AWS accounts that use consolidated billing. The company runs several active high performance Amazon RDS for Oracle On-Demand DB instances for 90 days. The company's finance team has access to AWS Trusted Advisor in the consolidated billing account and all other AWS accounts. The finance team needs to use the appropriate AWS account to access the Trusted Advisor check recommendations for RDS. The finance team must review the appropriate Trusted Advisor check to reduce RDS costs. Which combination of steps should the finance team take to meet these requirements?

- ☐ Use the Trusted Advisor recommendations from the account where the RDS instances are running.

Accessing Trusted Advisor from individual accounts, while possible, doesn't provide the consolidated view needed for a multi-account setup. It's more efficient to leverage the consolidated billing account for a holistic perspective.

- ☒ Use the Trusted Advisor recommendations from the consolidated billing account to see all RDS instance checks at the same time.

The consolidated billing account provides a centralized view of Trusted Advisor recommendations across all linked accounts, allowing the finance team to see all RDS instance checks in one place. This simplifies cost optimization efforts.

- ☐ Review the Trusted Advisor check for Amazon RDS Reserved Instance Optimization.

The Reserved Instance Optimization check assesses On-Demand RDS instances that are good candidates for Reserved Instances. This can lead to significant cost savings, particularly for instances running for extended periods.

- ☒ Review the Trusted Advisor check for Amazon RDS Idle DB Instances.

The Idle DB Instances check identifies instances that have not had connections for a prolonged period and suggests terminating them to reduce costs. Given the instances were active for the last 90 days, it might not be as effective as checking Reserved Instance Optimization, but is still useful in general for reducing costs.

- ☐ Review the Trusted Advisor check for Amazon Redshift Reserved Node Optimization.

This check is specific to Amazon Redshift, not RDS, and is therefore not relevant to optimizing RDS costs.

Overall Explanation:

To meet the requirements, the finance team should use Trusted Advisor from the consolidated billing account to see all RDS instance checks in one place. Also, reviewing the Amazon RDS Idle DB Instances check will help reduce costs for unused instances. While

the instances have been active for 90 days, it is still important to ensure that they are not currently idle. The Reserved Instance Optimization check helps to identify whether On-Demand RDS instances are good candidates to be converted to Reserved Instances, however due to the active instances it would be less helpful.

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Question 59

Not Attempted

Storage



A solutions architect needs to optimize storage costs for an application heavily reliant on Amazon S3. The architect must identify any S3 buckets that are no longer being accessed or are infrequently accessed to determine if the data can be moved to a less expensive storage class or archived. The solutions architect needs to accomplish this goal with the LEAST operational overhead.

- ☒ Analyze bucket access patterns by using the S3 Storage Lens dashboard for advanced activity metrics.

S3 Storage Lens provides a single view of object storage usage and activity across accounts, including metrics to identify infrequently accessed and unused buckets, thus optimizing costs with minimal overhead.

- ☐ Analyze bucket access patterns by using the S3 dashboard in the AWS Management Console.

While the S3 dashboard provides some basic metrics, it lacks the advanced analytics and comprehensive view of S3 Storage Lens, requiring more manual effort for identifying infrequent access patterns.

- ☐ Turn on the Amazon CloudWatch BucketSizeBytes metric for buckets. Analyze bucket access patterns by using the metrics data with Amazon Athena.

Using CloudWatch BucketSizeBytes provides information on storage size, but requires significant setup and querying with Athena to analyze access patterns, resulting in high operational overhead.

- ☐ Turn on AWS CloudTrail for S3 object monitoring. Analyze bucket access patterns by using CloudTrail logs that are integrated with Amazon CloudWatch Logs.

Enabling CloudTrail and integrating with CloudWatch Logs provides detailed access logs but incurs significant operational overhead due to the complexity of analyzing large log volumes to determine access patterns.

Overall Explanation:

S3 Storage Lens provides a fully managed solution for analyzing S3 storage usage and activity, including identifying infrequently accessed buckets. This allows the solutions architect to optimize storage costs with the least operational overhead compared to other options that require manual configuration, log analysis, or custom querying. Therefore, using the S3 Storage Lens dashboard is the most efficient solution.

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Question 60

Not Attempted

[Management & Governance](#)

A company hosts its web application on AWS and wants to ensure all Amazon EC2 instances, Amazon RDS DB instances, and Amazon Redshift clusters are configured with appropriate tags. The company requires a solution that minimizes configuration and operational overhead for this check. Which solution best meets these requirements?

- ☒ **Use AWS Config rules to define and detect resources that are not properly tagged.**
AWS Config provides a managed service for assessing, auditing, and evaluating resource configurations. It allows defining rules to check for proper tagging and automatically detect non-compliant resources, minimizing manual effort and operational overhead.
- ☐ **Use Cost Explorer to display resources that are not properly tagged. Tag those resources manually.**
Cost Explorer primarily focuses on cost analysis and doesn't provide direct enforcement or automated detection of improperly tagged resources. Manual tagging is labor-intensive and prone to errors.
- ☐ **Write API calls to check all resources for proper tag allocation. Periodically run the code on an EC2 instance.**
While this approach is feasible, it requires custom code development, maintenance, and operational overhead for managing the EC2 instance and scheduling.
- ☐ **Write API calls to check all resources for proper tag allocation. Schedule an AWS Lambda function through Amazon CloudWatch to periodically run the code.**
This option is better than option C because it removes the maintenance of an EC2 instance. However, it still involves custom code development and management, making it less efficient than using a managed service like AWS Config.

Overall Explanation:

AWS Config is the optimal solution because it provides a managed service specifically designed for monitoring and evaluating the configuration of AWS resources. By using AWS Config rules, the company can define the required tags and automatically detect any resources that do not comply with these rules, thus minimizing the effort of configuring and operating the check. Options involving custom API calls and EC2 instances or Lambda functions require more manual effort and maintenance compared to the automated approach provided by AWS Config. Cost Explorer is not designed for tag compliance enforcement.

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Question 61

Not Attempted

Storage



A company sells datasets to customers who do research in artificial intelligence and machine learning (AI/ML). The datasets are large, formatted files that are stored in an Amazon S3 bucket in the us-east-1 Region. The company hosts a web application that the customers use to purchase access to a given dataset. The web application is deployed on multiple Amazon EC2 instances behind an Application Load Balancer. After a purchase is made, customers receive an S3 signed URL that allows access to the files. The customers are distributed across North America and Europe. The company wants to reduce the cost that is associated with data transfers and wants to maintain or improve performance. What should a solutions architect do to meet these requirements?

- ☐ **✗** Configure S3 Transfer Acceleration on the existing S3 bucket. Direct customer requests to the S3 Transfer Acceleration endpoint. Continue to use S3 signed URLs for access control.

S3 Transfer Acceleration focuses on accelerating uploads to S3, which may not necessarily improve the performance needed for serving datasets to customers. It doesn't address data transfer cost reduction for downloads.

- ☒ **✓** Deploy an Amazon CloudFront distribution with the existing S3 bucket as the origin. Direct customer requests to the CloudFront URL. Switch to CloudFront signed URLs for access control.

Amazon CloudFront caches content at edge locations, reducing the need for frequent data transfer from the S3 bucket, thus significantly lowering data transfer costs. CloudFront delivers content to

users from the nearest edge location, minimizing latency and improving performance. CloudFront Signed URLs provide secure access control.

- ⊗ Set up a second S3 bucket in the eu-central-1 Region with S3 Cross-Region Replication between the buckets. Direct customer requests to the closest Region. Continue to use S3 signed URLs for access control.

Setting up a second S3 bucket with cross-region replication increases the cost rather than reducing it. While it can improve latency for European customers, it introduces higher costs for data replication.

- ⊗ Modify the web application to enable streaming of the datasets to end users. Configure the web application to read the data from the existing S3 bucket. Implement access control directly in the application.

Modifying the web application to stream datasets is complex to implement and doesn't directly address data transfer cost reduction. Implementing access control in the application itself is also less secure and scalable than using signed URLs.

Overall Explanation:

To reduce the cost associated with data transfers and maintain or improve performance, deploying an Amazon CloudFront distribution with the existing S3 bucket as the origin is the best approach. CloudFront caches data at edge locations closer to the users, reducing data transfer costs and improving latency. CloudFront signed URLs provide secure access to the data.

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Question 62

Not Attempted

[Management & Governance](#)

A company is designing a web application on AWS to process insurance quotes. Users will request quotes through the application. These quotes must be categorized by quote type, responded to within 24 hours, and must not be lost. The solution should maximize operational efficiency and minimize maintenance overhead. Which solution best meets these requirements?

- ⊗ Create multiple Amazon Kinesis data streams based on the quote type. Configure the web application to send messages to the proper data stream. Configure each backend group of application servers to use the Kinesis Client Library (KCL) to poll messages from its own data stream.

This option involves managing multiple Kinesis streams and requires more complex KCL configuration, increasing operational overhead and maintenance.

- ⊗ Create an AWS Lambda function and an Amazon Simple Notification Service (Amazon SNS) topic for each quote type. Subscribe the Lambda function to its associated SNS topic. Configure the application to publish requests for quotes to the appropriate SNS topic.

Creating multiple SNS topics and Lambda functions, one for each quote type, does not maximize operational efficiency and can become difficult to manage.

- ✔ Create a single Amazon Simple Notification Service (Amazon SNS) topic. Subscribe Amazon Simple Queue Service (Amazon SQS) queues to the SNS topic. Configure SNS message filtering to publish messages to the proper SQS queue based on the quote type. Configure each backend application server to use its own SQS queue.

This option leverages SNS fan-out with SQS queues and message filtering, enabling efficient routing and reliable queuing while minimizing maintenance using managed services.

- ⊗ Create multiple Amazon Kinesis Data Firehose delivery streams based on the quote type to deliver data streams to an Amazon OpenSearch Service cluster. Configure the application to send messages to the proper delivery stream. Configure each backend group of application servers to search for the messages from OpenSearch Service and process them accordingly.

Using Kinesis Data Firehose and OpenSearch Service for this purpose is an over-engineered solution. OpenSearch is better suited for analytics/logging rather than transactional quote processing, and adds unnecessary complexity.



The best solution is to use a single SNS topic with multiple SQS queues, leveraging SNS message filtering to route messages based on quote type. SQS ensures reliable message delivery and prevents data loss, while SNS simplifies message distribution. This setup is efficient, scalable, and minimizes maintenance by utilizing fully managed AWS services. This architecture is known as the SNS fan-out pattern. SQS guarantees delivery within 24 hours by default, and the system is built around queues, so no messages will be lost.

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Question 63

Not Attempted

[Management & Governance](#)

A company's application runs on several Amazon EC2 instances, each with multiple Amazon Elastic Block Store (Amazon EBS) data volumes attached. The company needs a nightly backup solution that protects both the EC2 instance configuration and data. Furthermore, the application must be recoverable in a different AWS Region. Which solution meets these requirements in the MOST operationally efficient way?

- ☐ Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Region.

This option requires custom coding and management of Lambda functions, which increases operational overhead. It also focuses solely on EBS volumes and may not fully capture the EC2 instance configuration.

- ☒ Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EC2 instances as resources.

This is the most operationally efficient solution. AWS Backup provides a centralized and automated backup management service. Backing up the EC2 instances also backs up the associated EBS volumes and the EC2 instance configurations, and facilitates cross-region recovery.

- ☐ Create a backup plan by using AWS Backup to perform nightly backups. Copy the backups to another Region. Add the application's EBS volumes as resources.

This option only backs up the EBS volumes and not the EC2 instance configurations. Recovering the application would be more complex as it wouldn't have EC2 instance configurations.

- ☐ Write an AWS Lambda function that schedules nightly snapshots of the application's EBS volumes and copies the snapshots to a different Availability Zone.

This option only provides high availability within the same Region and does not address the requirement of disaster recovery in a different Region. It also requires custom coding and management.

Overall Explanation:

The most operationally efficient solution is to use AWS Backup and add the EC2 instances as resources in the backup plan. This approach automatically backs up both the EC2 configuration and the attached EBS volumes. AWS Backup also provides native support for copying backups to another Region for disaster recovery. Options involving custom Lambda functions require more manual configuration and management overhead. Backing up only the EBS volumes without EC2 instances complicates recovery of the application's full configuration.

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Question 64

Not Attempted

Security



A company is building a mobile application on AWS. The company aims to expand its reach to millions of users. To achieve this, the company needs to create a platform that allows authorized users to watch the company's content on their mobile devices securely and efficiently. Which solution should a solutions architect recommend to meet these requirements?

- ☐ Publish content to a public Amazon S3 bucket. Use AWS Key Management Service (AWS KMS) keys to stream content.

Publishing content to a public S3 bucket does not provide sufficient control over access and security. While KMS can encrypt the data at rest, it doesn't effectively manage access during streaming. This is not a scalable or secure approach for millions of users.

- ☐ Set up IPsec VPN between the mobile app and the AWS environment to stream content.

Setting up an IPsec VPN for each mobile user is not a scalable or practical solution for millions of users. The overhead of managing and maintaining numerous VPN connections would be significant, and it would negatively impact performance.

- ☒ Use Amazon CloudFront. Provide signed URLs to stream content.

Amazon CloudFront is a content delivery network (CDN) that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds. CloudFront supports signed URLs that provide authorized access to your content. This feature allows the company to control who can access their content and for how long, providing a secure and scalable solution for millions of users.

- ☐ Set up AWS Client VPN between the mobile app and the AWS environment to stream content.

AWS Client VPN is designed for secure remote access to internal resources, not for streaming content to a large user base. It is not a scalable or cost-effective solution for millions of mobile users.

Overall Explanation:

Amazon CloudFront is the most suitable solution because it is a CDN that securely delivers content globally with low latency and high transfer speeds. Signed URLs provide

authorized access control, ensuring that only authenticated users can stream the company's content. This approach is highly scalable and efficient for millions of users.

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Question 65

Not Attempted

Database



A company has an on-premises MySQL database used by the global sales team with infrequent access patterns. The sales team requires the database to have minimal downtime. A database administrator wants to migrate this database to AWS without selecting a particular instance type in anticipation of more users in the future. Which service should a solutions architect recommend?

☐ Amazon Aurora MySQL

While Aurora MySQL offers high performance and availability, it requires selecting a specific instance type, which the administrator wants to avoid.

☒ Amazon Aurora Serverless for MySQL

Aurora Serverless for MySQL automatically scales capacity based on application needs without requiring instance type selection, making it ideal for infrequent access patterns and future growth. It also provides minimal downtime.

☐ Amazon Redshift Spectrum

Amazon Redshift Spectrum is designed for querying data in Amazon S3 data lakes and is not suitable for migrating an on-premises MySQL database.

☐ Amazon RDS for MySQL

RDS for MySQL requires selecting a specific instance type and managing its scaling, which contradicts the requirement of avoiding instance type selection.

Overall Explanation:

Amazon Aurora Serverless for MySQL is the correct choice because it eliminates the need to select a specific instance type, automatically scales based on demand, and provides minimal downtime, aligning with the requirements of infrequent access patterns and future growth. RDS for MySQL and Aurora MySQL require instance selection. Redshift Spectrum is for querying S3 data.

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